# AI POWERED ACCESSIBILITY FOR ENABLING EFFECTIVE COMMUNICATION FOR HEARING AND SPEECH IMPAIRED IN VIRTUAL PLATFORMS

**A PROJECT REPORT**

***Submitted by***


| | |
|---|---|
| **DINESH KUMAR G** | **-510921104009** |
| **MOHAMMAD SHAMIL** | **-510921104028** |
| **RAASHID A** | **-510921104040** |
| **SANTHOSH V** | **-510921104047** |


***In partial fulfilment of the award of the degree***

***Of***

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## ANNA UNIVERSITY: CHENNAI 600 025

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# ANNA UNIVERSITY: CHENNAI 600 025

## BONAFIDE CERTIFICATE

This is to certify that the project title **"AI Powered Accessibility for Enabling Effective Communication for Hearing and Speech Impaired in Virtual Platforms"** is the Bonafide work of **DINESH KUMAR. G [510921104009], MOHAMMED SHAMIL [510921104028], RAASHID. A [510921104040], SANTHOSH. V [510921104047]** who carried out the project work. Certified further, the work reported here in does not form a part of any other project works on the basis of which a degree or award was conferred earlier on this title.

| **SIGNATURE** | **SIGNATURE** |
|---|---|
| Dr. J. DAPHNEY JOANN, M.E., PhD., | Dr. D. UDAYA SURIYA RAJKUMAR, M.E(CSE)., PhD., |
| **SUPERVISOR** | **HEAD OF THE DEPARTMENT** |
| Assistant Professor | Professor |
| Artificial Intelligence and Data Science | Computer Science and Engineering |
| Global Institute of Engineering and Technology, Ranipet - 632 509 | Global Institute of Engineering and Technology, Ranipet - 632 509 |

Submitted for project viva voce examination held on ……

**INTERNAL EXAMINER** **EXTERNAL EXAMINER**

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**Signature of the Candidate**

## ABSTRACT

Effective communication is vital for sharing information, ideas, and emotions. However, virtual meeting platforms like Zoom, Microsoft Teams, and Google Meet often fail to accommodate individuals with hearing and speech impairments, creating significant barriers to inclusivity. While sign language offers a means of communication for deaf individuals, its interpretation remains challenging for non-signers. Existing sign language recognition technologies are limited in accuracy and accessibility, making them unsuitable for seamless integration into virtual platforms. This project introduces an AI-driven system designed to bridge the communication gap between deaf and hearing participants in virtual meetings. The system employs advancements in deep learning, particularly Temporal Convolutional Networks (TCNs), to enable two-way communication in real-time. It includes three core modules: a Sign Recognition Module (SRM) that interprets signs using TCN, a Speech Recognition and Synthesis Module (SRSM) powered by Hidden Markov Models, which converts spoken language into text, and an Avatar Module (AM) that visually translates speech into corresponding signs. The Avatar Module is essential for visually representing spoken language in sign language format, ensuring non-signers can effectively communicate with sign language users in an intuitive and engaging way. This solution represents a significant advancement in fostering inclusivity and accessibility in virtual meeting environments.

# TABLE OF CONTENTS


| CHAPTER. NO | TITLE | PAGE NO |
|---|---|---|
| | **ABSTRACT** | **IV** |
| | **LIST OF FIGURES** | **VII** |
| | **LIST OF ABBREVIATIONS** | **VIII** |
| **1** | **INTRODUCTION** | **01** |
| | 1.1. PROBLEM STATEMENT | **04** |
| | 1.2. AIM AND OBJECTIVE | **04** |
| | 1.3. SCOPE OF THE PROJECT | **05** |
| **2** | **LITRATURE SURVEY** | **06** |
| **3** | **EXISTING SYSTEM** | **10** |
| | 3.1. DISADVANTAGES | **11** |
| **4** | **PROPOSED SYSTEM** | **12** |
| | 4.1 ADVANTAGES | **13** |
| **5** | **SYSTEM SPECIFICATION** | **14** |
| | 5.1. HARDWARE REQUIREMENTS | **14** |
| | 5.2. SOFTWARE REQUIREMENTS | **14** |
| **6** | **SYSTEM DESIGN** | **17** |
| | 6.1. SYSTEM ARCHITECTURE | **17** |
| | 6.2. DATA FLOW DIAGRAM | **18** |
| | 6.3. UML DIAGRAM | **21** |
| **7** | **SYSTEM IMPLEMENTATION** | **27** |
| | 7.1. SYSTEM DESCRIPTION | **28** |
| | 7.2. MODULE DESCRIPTION | **30** |
| **8** | **TESTING** | **35** |
| | 8.1. SOFTWARE TESTING | **35** |
| | 8.2. TEST CASES | **36** |

| | | |
|---|---|---|
| | 8.3. TEST REPORT | **37** |
| **9** | **PERFORMANCE ANALYSIS** | **38** |
| **10** | **CONCLUSION AND FUTURE SCOPE** | **40** |
| | APPENDIX 1: SAMPLE CODINGS | **42** |
| | APPENDIX 2: SNAP SHOTS | **46** |
| | **REFERENCES** | **51** |

## LIST OF FIGURES


| FIGURE NUMBER | FIGURES | PAGE NUMBER |
|---|---|---|
| 1.1 | Sign Language | 01 |
| 1.2 | Deep Neuron | 02 |
| 1.3 | Convolution Layer | 03 |
| 1.4 | Temporal Convolutional Networks | 04 |
| 6.1 | System Architecture | 17 |
| 6.2.1 | Data Flow Level 0 | 18 |
| 6.2.2 | Data Flow Level 1 | 19 |
| 6.2.3 | Data Flow Level 2 | 20 |
| 6.3.1 | Use Case Diagram | 21 |
| 6.3.2 | Class Diagram | 22 |
| 6.3.3 | Activity Diagram | 23 |
| 6.3.4 | Sequence Diagram | 24 |
| 6.3.5 | Collaboration Diagram | 25 |
| 6.3.6 | Component Diagram | 26 |

## LIST OF ABBREVIATIONS

| S.NO | ABBREVIATIONS | DESCRIPTION |
|---|---|---|
| 1 | TCNS | Temporal Convolutional Networks |
| 2 | SRM | Sign Recognition Module |
| 3 | SRSM | Speech Recognition and Synthesis Module |
| 4 | AM | Avatar Module |
| 5 | SVM | Support Vector Machines |
| 6 | RPN | Region Proposal Network |
| 7 | HMM | Hidden Markov Models |

# CHAPTER 1
# INTRODUCTION

Sign language is manual communication commonly used by people who are deaf. Sign language is not universal; people who are deaf from different countries speak different sign languages. The gestures or symbols in sign language are organized in a linguistic way. Each individual gesture is called a sign. Each sign has three distinct parts: the handshape, the position of the hands, and the movement of the hands. American Sign Language (ASL) is the most commonly used sign language in the India.

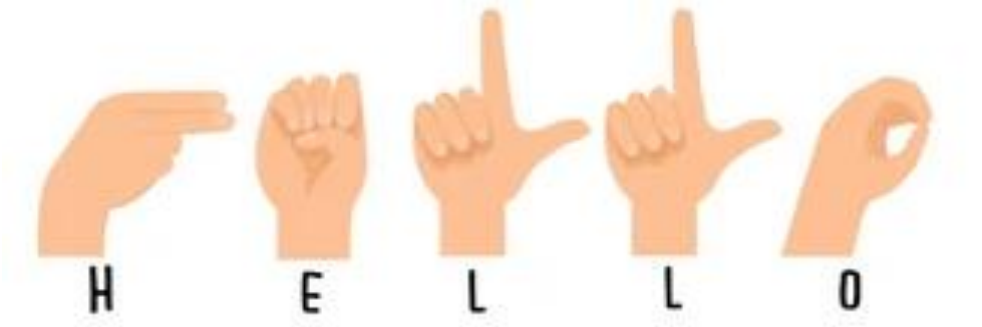



**Fig 1.1. Sign Language**

A sign language (also signed language) is a language which uses manual communication, body language, and lip patterns instead of sound to convey meaning—simultaneously combining hand shapes, orientation and movement of the hands, arms or body, and facial expressions to fluidly express a speaker's thoughts. Signs often represent complete ideas, not only words. However, in addition to accepted gestures, mime, and hand signs, sign language often includes finger spelling, which involves the use of hand positions to represent the letters of the alphabet. Sign languages have developed in circumstances where groups of people with mutually unintelligible spoken languages found a common base and were able to develop signed forms of communication. Sign languages commonly develop in deaf communities, which include people who are deaf or hard of hearing, friends and families of deaf people, as well as interpreters.

## Deep Learning

Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is inspired by the human brain. Deep learning models can recognize complex patterns in pictures, text, sounds, and other data to produce accurate insights and predictions.

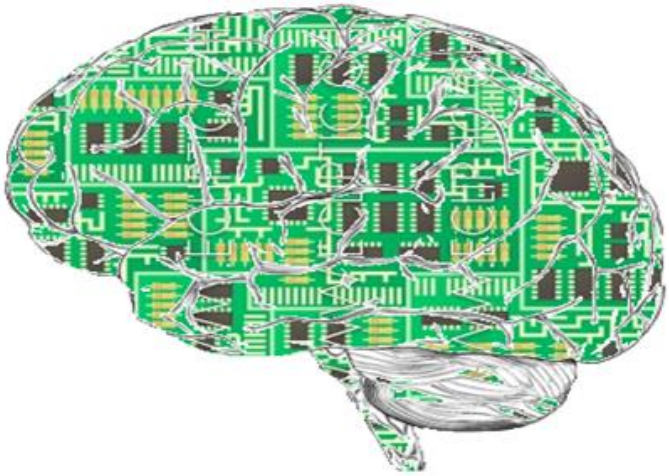


**Fig 1.2: Deep Neuron**

Deep learning algorithms are neural networks that are modeled after the human brain. For example, a human brain contains millions of interconnected neurons that work together to learn and process information. Similarly, deep learning neural networks, or artificial neural networks, are made of many layers of artificial neurons that work together inside the computer.

## Convolutional Neural Networks (CNN)

In CNN, the processing of data involves breaking the images into many numbers of overlapping tiles instead of feeding entire images into our network. And then, we use a technique called a sliding window over the whole original image and save the results as a separate tiny picture tile. The Sliding window is a kind of brute force solution where we scan all around for a given image to detect the object for all possible sections, each section at a time until we get the expected object.

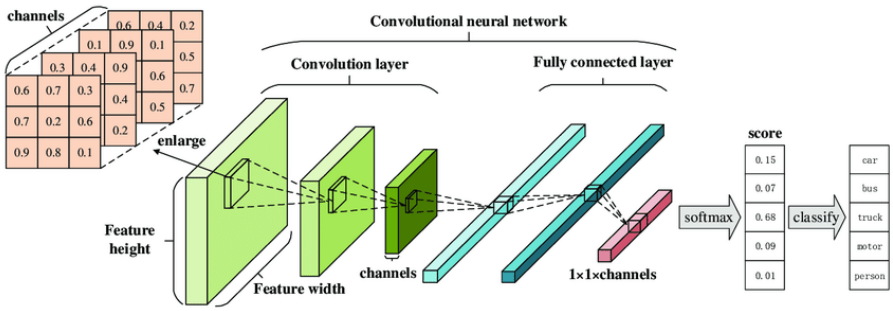



**Fig 1.3. Convolution Layer**

### Convolution Layer

CNN has a convolution layer that has several filters to perform the convolution operation.

### Rectified Linear Unit (ReLU)

CNN's have a ReLU layer to perform operations on elements. The output is a rectified feature map.

### Pooling Layer

The rectified feature map next feeds into a pooling layer. Pooling is a down-sampling operation that reduces the dimensions of the feature map. The pooling layer then converts the resulting two-dimensional arrays from the pooled feature map into a single, long, continuous, linear vector by flattening it.

### Temporal Convolutional Networks

The seminal work of Lea et al. (2016) first proposed a Temporal Convolutional Networks (TCNs) for video-based action segmentation. The two steps of this conventional process include: firstly, computing of low-level features using (usually) CNN that encode spatial-temporal information and secondly, input these low-level features into a classifier that captures high-level temporal information using (usually) RNN. The main disadvantage of such an approach is that it requires two separate models. TCN provides a unified approach to capture all two levels of information hierarchically.

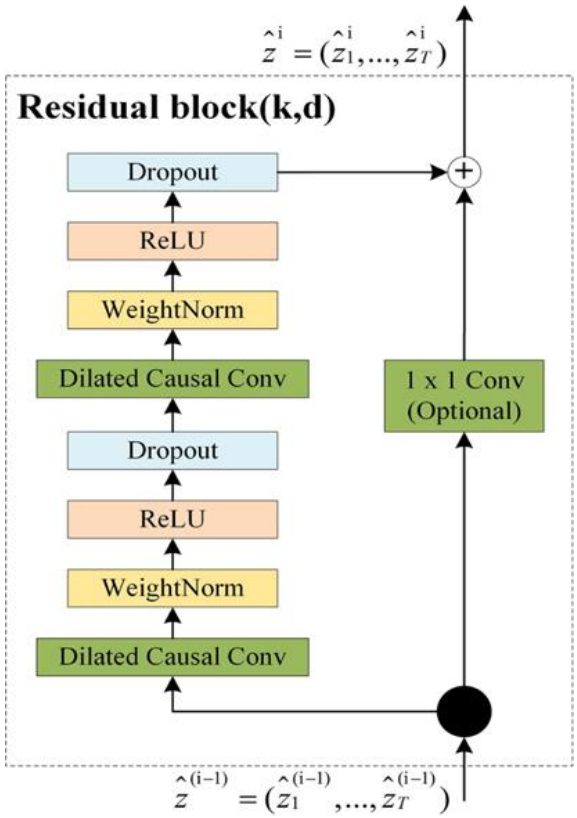



**Fig 1.4. Temporal Convolutional Networks**

## 1.1. PROBLEM STATEMENT

In today's digital era, virtual meeting platforms like Zoom, Microsoft Teams, and Google Meet have become essential for communication and collaboration. However, these platforms often lack effective accessibility features for deaf and non-deaf users to interact seamlessly. Deaf individuals face significant challenges in understanding spoken content during virtual meetings, while non-deaf users struggle to communicate with their deaf counterparts due to the absence of real-time sign language interpretation.

## 1.2. AIM AND OBJECTIVE

The aim of the project is to develop an AI-powered system that facilitates seamless communication between hearing and speech-impaired individuals and hearing users by integrating sign language recognition, speech-to-text conversion, and visual representation in virtual platforms that will seamlessly connect the signers to non-signers.

- To develop a Sign Recognition Module that translates sign language into text or speech.
- To create a Speech Recognition and Synthesis Module that converts spoken language into text.
- To implement an Avatar Module for visual sign language representation.
- To ensure real-time communication in virtual platforms like Zoom, Microsoft Teams, and Google Meet.
- To enhance accessibility and inclusivity for individuals with hearing and speech impairments.

## 1.3. SCOPE OF THE PROJECT

This project focuses on enhancing communication accessibility by integrating AI-driven sign language recognition and speech-to-text conversion into popular virtual meeting platforms. It aims to support real-time, inclusive interactions for diverse users, including those with hearing and speech impairments.

- **Inclusive Communication**: Bridges the gap between hearing and speech-impaired individuals and hearing users through AI-driven sign language recognition and speech-to-text conversion.
- **Virtual Platform Integration**: Compatible with popular platforms like Zoom, Microsoft Teams, and Google Meet to ensure accessibility in virtual meetings.
- **Real-Time Translation**: Provides instant sign-to-text, speech-to-text, and text-to-sign conversion for smooth interaction.
- **Multi-User Support**: Beneficial for deaf, hard-of-hearing, mute individuals, and non-signers, promoting effective communication.
- **AI-Driven Accuracy**: Uses **Temporal Convolutional Networks (TCN)** and **Hidden Markov Models (HMM)** to improve recognition accuracy.

# CHAPTER 2

# LITERATURE SURVEY

1. **Title**: **Hand and Pose-Based Feature Selection for Zero-Shot Sign Language Recognition**.

   **Authors:** Giray Sercan Özcan; Yunus Can Bilge **Year:** 2024

   **Link:** https://ieeexplore.ieee.org/document/10620240

   **Objective**

   To improve the accuracy and generalization of Zero-Shot Sign Language Recognition by leveraging a combination of hand, pose, and body landmark data, enabling the recognition of unseen sign classes through enhanced visual and semantic feature alignment**.**

   **Summary**

   This study presents a Zero-Shot Sign Language Recognition framework that combines hand, pose, and full-body landmark data for enhanced visual representation. The method employs a data grading mechanism and self-attention to select and refine features, improving the semantic alignment between visual inputs and textual embeddings. It incorporates several advanced algorithms, including ST-GCN, MViTv2, ResNeXt, Bi-LSTM, and CLIP architecture. Evaluations on MS-ZSSLR-W, MS-ZSSLR-C, and ASL-Text datasets demonstrate improved performance over previous ZSSLR methods, addressing the challenge of recognizing signs not seen during training.

2. **Title: Hybrid InceptionNet Based Enhanced Architecture for Isolated Sign Language Recognition**

   **Authors:** Deep R. Kothadiya; Chintan M. Bhatt **Year**: 2024

   **Link:** https://ieeexplore.ieee.org/document/10577129

   **Objective**

   To enhance the recognition accuracy of isolated sign language gestures using a

deep learning architecture that combines multiple convolutional neural networks within a hybrid Inception framework.

**Summary**

This study introduces a hybrid deep learning model based on an enhanced InceptionNet architecture aimed at improving the classification of isolated sign language gestures. The approach integrates multiple convolutional neural networks in an ensemble setup, leveraging InceptionV4 and optimized backpropagation to strengthen feature extraction and model performance. Evaluated on a benchmark dataset for isolated sign language gestures, the model achieves a high accuracy of 98.46%. The architecture is designed to capture complex gesture variations, although its increased computational complexity and model size may hinder real-time deployment.

3. **Title**: **Deep Learning-Based Standard Sign Language Discrimination**

   **Authors:** Menglin Zhang; Shuying Yang **Year:** 2023

   **Link:** https://ieeexplore.ieee.org/document/10310202

   **Objective**

   To accurately distinguish between standard and non-standard sign language actions within educational software, supporting more effective sign language learning and assessment.

   **Summary**

   This study proposes a deep learning-based system designed to assess the correctness of sign language gestures in educational settings. It employs flow-guided features for precise hand detection using a Faster RCNN with Feature Pyramid Network (FPN), and utilizes an encoder-decoder model with both 2D and 3D convolutional layers to capture temporal and spatial dependencies in gesture sequences. Tested on the SLCD dataset, the model demonstrates high hand detection accuracy and reduced computational cost, making it suitable for interactive learning environments. However, its use is mainly confined to

educational scenarios, limiting broader application in real-world sign language recognition or translation tasks.

4. **Title: Sign4all: A Low-Cost Application for Deaf People Communication**
   **Author:** Francisco Morillas-Espejo **Year:** 2023
   **Link:** https://ieeexplore.ieee.org/document/10242052
   **Objective**
   To provide an accessible and low-cost solution that enables communication between deaf and hearing individuals by recognizing the sign language alphabet.
   **Summary**
   This study presents Sign4all, an application designed to bridge communication gaps between deaf and hearing individuals by recognizing signs from the Spanish Sign Language (LSE) alphabet. The system utilizes a convolutional neural network based on ResNet50, along with post-processing techniques, to ensure accurate classification of alphabet signs. Recognized signs are then translated and displayed through a virtual avatar, enhancing user interaction. The application's low cost and practical design make it well-suited for everyday use. However, its functionality is limited to alphabet recognition, reducing its utility in more advanced or natural conversational scenarios.

5. **Title:** An Efficient Two-Stream Network for Isolated Sign Language Recognition Using Accumulative Video Motion
   **Author:** Hamzah Luqman **Year:** 2022
   **Link:** https://ieeexplore.ieee.org/document/9875269
   **Objective**
   To develop an efficient deep learning framework for recognizing isolated sign language gestures by capturing and refining motion information from video sequences.
   **Summary**
   This study introduces a two-stream neural network architecture designed to

improve the recognition of isolated sign language gestures through motion analysis. The model integrates three distinct streams: Deep Motion Network (DMN), Accumulative Motion Network (AMN), and Sequence Refinement Network (SRN), enabling effective motion feature extraction and sequence-level refinement. Tested on the KArSL-190 and KArSL-502 datasets, the approach demonstrates high recognition efficiency and accuracy. However, the datasets used suffer from limited signer diversity, which may impact the model's generalization to broader user populations.

6. **Title: A Wearable Smart Glove and Its Application of Pose and Gesture Detection to Sign Language Classification**

   **Authors:** Joseph DelPreto; Josie Hughes **Year: 2022**

   **Link:** https://ieeexplore.ieee.org/document/9830849

   **Objective**

   To design and implement a real-time, wearable solution for the classification of American Sign Language (ASL) poses and gestures, using a smart glove equipped with soft sensors and an accelerometer. The goal is to create a portable system capable of accurately detecting and interpreting hand movements and gestures as part of an assistive communication tool for individuals who rely on ASL.

   **Summary**

   This study presents a wearable smart glove system designed for real-time ASL pose and gesture classification. The system integrates soft sensors and an accelerometer with a Long Short-Term Memory (LSTM) neural network, enabling accurate detection and interpretation of hand movements. By capturing fine-grained hand gestures and poses, the glove provides a portable solution for ASL recognition, useful for communication assistance. However, the system's performance is dependent on user-specific calibration, as variations in hand size, finger flexibility.

# CHAPTER 3
# EXISTING SYSTEM

Most virtual communication platforms like Zoom, Microsoft Teams, and Google Meet do not offer effective real-time sign language interpretation. This creates barriers for individuals with hearing impairments, making it difficult for them to actively participate in virtual meetings, discussions, and conferences. Here the deaf users are neglected, so there is a loss of communication in-between deaf and non-deaf users.

- **Assistive Devices**

Traditional assistive devices, such as hearing aids and cochlear implants, are designed to improve auditory perception for individuals with hearing loss. However, these devices do not address the communication challenges faced by those who rely on sign language, making them an incomplete solution for inclusive communication.

- **Manual Communication Tools**

Basic manual communication methods, such as pen and paper, are often used by individuals with hearing and speech impairments to convey messages. While effective for simple conversations, this approach is slow, inconvenient, and impractical for dynamic interactions in digital and professional environments.

- **Sign Language Interpreters**

In various settings, human sign language interpreters help bridge the communication gap between deaf and hearing individuals. However, interpreters are not always available, can be costly, and require prior arrangements, making them less practical for spontaneous or everyday conversations. Human sign language interpreters play a vital role in facilitating communication between deaf and hearing individuals across different settings. Their availability is often limited, they can be expensive to hire, and they usually require advance scheduling, making them impractical for spontaneous or everyday interactions.

## 3.1. DISADVANTAGES

- Virtual platforms lack built-in real-time sign language recognition and translation.
- Devices like cochlear implants do not support sign language users.
- Writing-based communication is slow and impractical for dynamic conversations.
- Sign language interpreters are not always accessible and can be costly.
- Struggles with large datasets and continuous sign sequences, reducing real-time effectiveness.
- Ensemble learning models require significant resources for training and deployment.
- GRBP-based methods rely on high-quality skeletal tracking, which may not be feasible in all environments.
- High dependency on specialized hardware (e.g., depth sensors, high-resolution cameras) limits the accessibility and scalability of some recognition systems.
- Latency issues in real-time AI processing can disrupt the flow of communication, especially in professional or time-sensitive contexts.
- Ensemble learning models require significant resources for training and deployment.
- Frequent recalibration needed for wearable devices (like smart gloves) affects user convenience.
- Real-time systems often struggle with background noise, occlusions, or low-light conditions during video-based recognition.
- Lack of standardized datasets across different sign languages hampers the development of universally adaptable models.
- Limited multilingual support restricts cross-regional accessibility, as many existing systems are tailored to a single sign language.

# CHAPTER 4
# PROPOSED SYSTEM

The proposed system introduces an AI-driven sign language interpretation framework that enhances accessibility for individuals with hearing and speech impairments. By integrating deep learning and natural language processing techniques, the system enables seamless two-way communication between sign language users and non-signers.

- **Sign Recognition Module (SRM)**

The Sign Recognition Module (SRM) utilizes Temporal Convolutional Networks (TCNs) to interpret sign language gestures accurately from live video input. This module processes hand movements, facial expressions, and body posture to ensure precise recognition of signs.

- **Speech Recognition and Synthesis Module (SRSM)**

The Speech Recognition and Synthesis Module (SRSM) is responsible for converting spoken language into text and vice versa. This module employs Hidden Markov Models (HMMs) to process speech signals, transcribe them into text, and generate synthesized speech when needed. This feature allows users who rely on speech-based communication to effectively interact with sign language users, thereby bridging the communication gap.

- **Avatar Module (AM)**

The Avatar Module (AM) enhances accessibility by generating a real-time animated avatar that visually translates spoken language into sign language. This module ensures that deaf users who prefer sign language can receive information in an intuitive and interactive manner. The avatar mimics human sign language gestures.

- **Integration with Virtual Platforms**

The system is designed to be seamlessly integrated with popular virtual meeting platforms such as Zoom, Microsoft Teams, and Google Meet. This integration

ensures that individuals with hearing and speech impairments can participate fully in discussions without communication barriers. The system operates in real time, providing accurate translations and ensuring that sign language users are not excluded from professional or social interactions.

## 4.1. ADVANTAGES

- Enhances communication between sign language users and non-signers.
- Provides real-time sign language interpretation for virtual meetings.
- Ensures high accuracy using deep learning-based recognition models.
- Eliminates the need for human interpreters in digital communication.
- Works effectively in noisy environments where speech recognition alone may fail.
- Promotes inclusivity for individuals with hearing and speech impairments.
- Facilitates effective participation in virtual meetings and online interactions.
- Reduces dependency on expensive assistive communication tools.
- Supports multi-user interactions in diverse communication settings.
- Contributes to the advancement of AI-driven assistive technology solutions.
- Provides an intuitive, user-friendly interface for both sign language users and non-signers.
- Supports continuous learning and improvement of sign language models through machine learning techniques.
- Can be customized to support multiple sign languages, enhancing global accessibility.
- Improves user engagement and comprehension through the use of lifelike avatars, which offer visual clarity and emotional expression that enhance the delivery of nuanced messages in sign language.

# CHAPTER 5

# SYSTEM SPECIFICATION

## 5.1. HARDWARE REQUIREMENTS

- **Processor:** Intel Core i5 or higher
- **RAM:** 8GB or more
- **Storage:** 256GB SSD or higher
- **Camera & Microphone:** For real-time gesture and speech recognition

## 5.2. SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10 or 11 (for Windows-specific development)
- **Programming Language:** Python (version 3.6 or higher)
- **Web Framework:** Flask for implementing a web-based user interface,
- **Database Integration:** MySQL for storing and retrieving relevant data.
- **Integrated Development Environment (IDE):** IDLE
- **Web Technologies:** HTML, CSS, and JavaScript, Bootstrap
- **Libraries & Packages:** TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Pillow, Scikit-learn.
- **Video Calling Tools**: jitsi-For enabling real-time video calls.

## SOFTWARE DESCRIPTION

### PYTHON 3.8

Python is a high-level, interpreted programming language known for its emphasis on code readability, simplicity, and versatility. Designed by Guido van Rossum and first released in 1991, Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming. It features dynamic typing, automatic memory management, and an extensive standard library, making it well-suited for rapid application development .




**Tensor Flow**

TensorFlow is an end-to-end, open-source machine learning platform developed by the Google Brain team. It provides a comprehensive suite of tools, libraries, and community-supported resources for designing, building, training, and deploying machine learning and deep learning models at scale.




**Keras**

**Keras** is an open-source, high-level deep learning API in Python, designed for building, training, and evaluating neural networks with ease and flexibility. Developed by François Chollet and now integrated into TensorFlow, it supports rapid prototyping while offering powerful computational tools through a simple, modular interface.



**Pandas**

Pandas is an open-source data analysis and manipulation library for the Python programming language, providing high-performance, user-friendly data structures and functions specifically designed for working with structured data. Built on top of NumPy, Pandas introduces the DataFrame and Series objects, enabling efficient handling of tabular and labeled data.




**NumPy**

NumPy (Numerical Python) is a fundamental open-source library for numerical computing in Python. It provides the ndarray, a high-performance multidimensional array object, along with a wide range of mathematical functions for efficient operations on large numerical datasets.




**Matplotlib**

Matplotlib is a widely-used open-source Python library for creating static, animated, and interactive visualizations. It offers a versatile set of tools for generating high-quality plots, charts, and graphs, making it essential for data analysis, scientific research, and reporting.




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# CHAPTER 6

# SYSTEM DESIGN

## 6.1. SYSTEM ARCHITECTURE

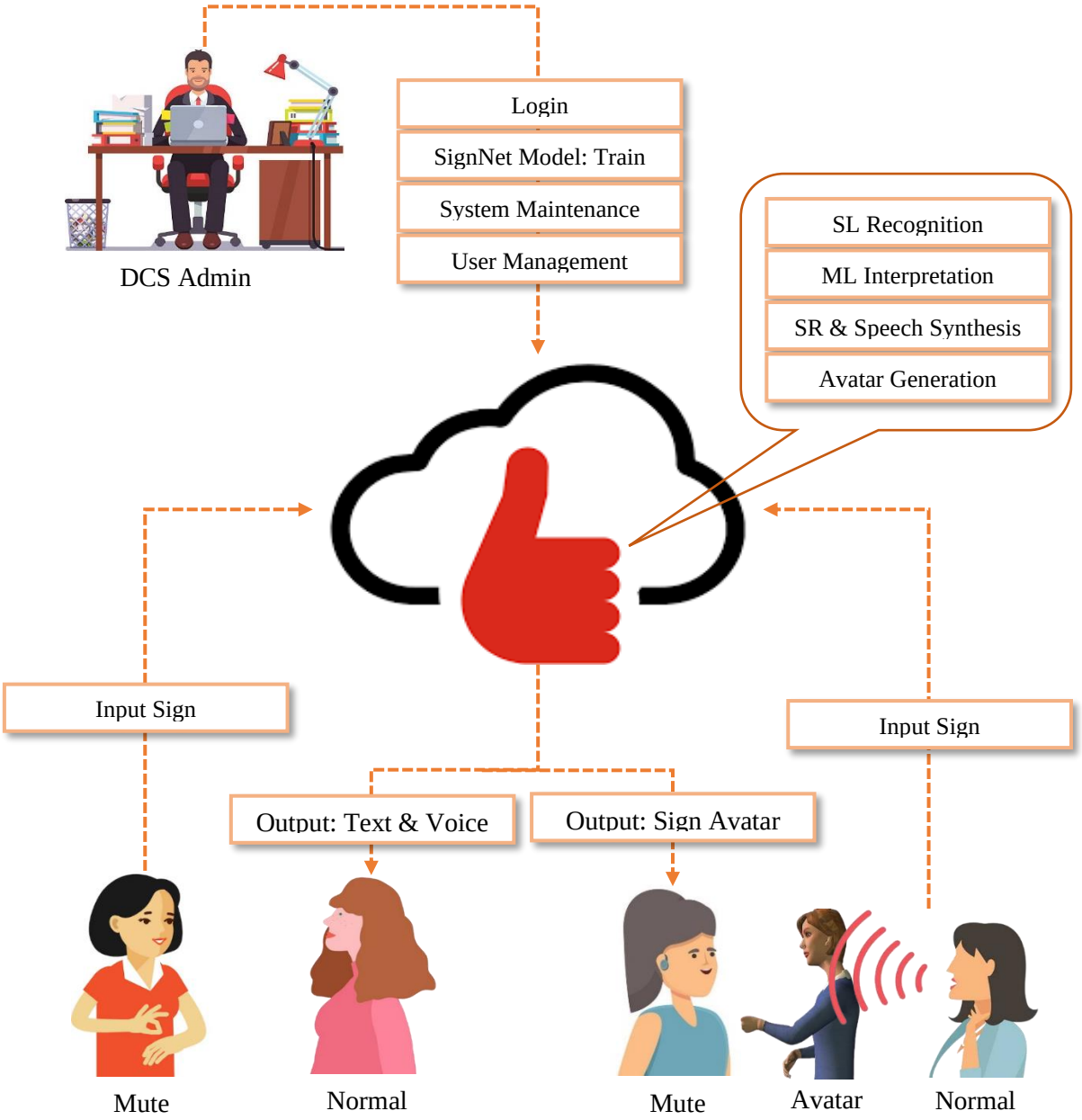



**Fig 6.1. System Architecture**

## 6.2. DATA FLOW DIAGRAM

## LEVEL 0

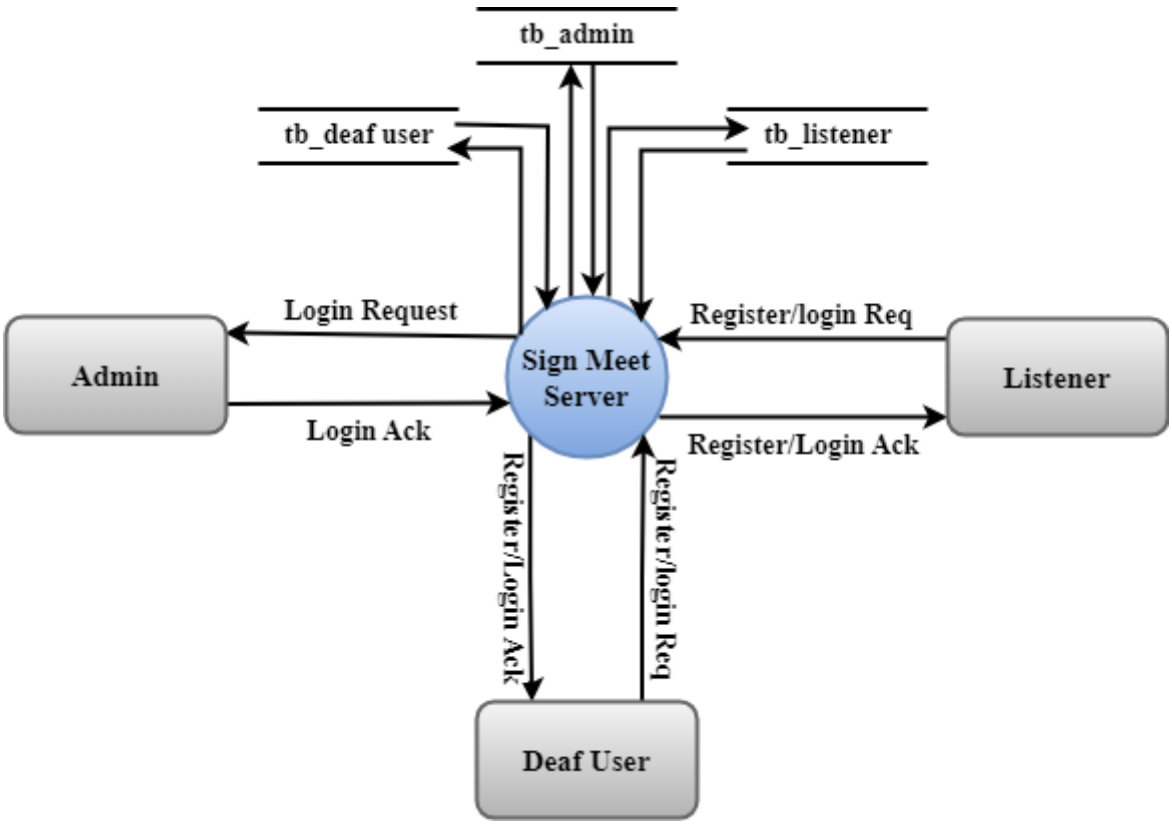



**Fig 6.2.1. Data Flow Level 0**

**LEVEL 1**

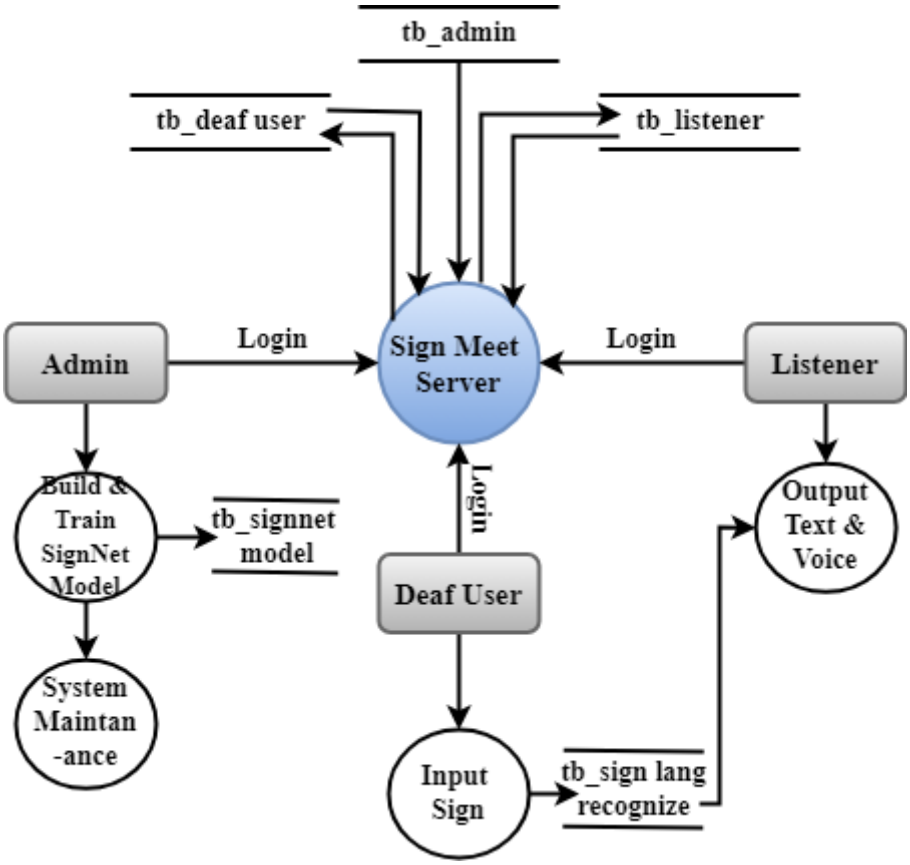



**Fig 6.2.2. Data Flow Level 1**

## LEVEL 2

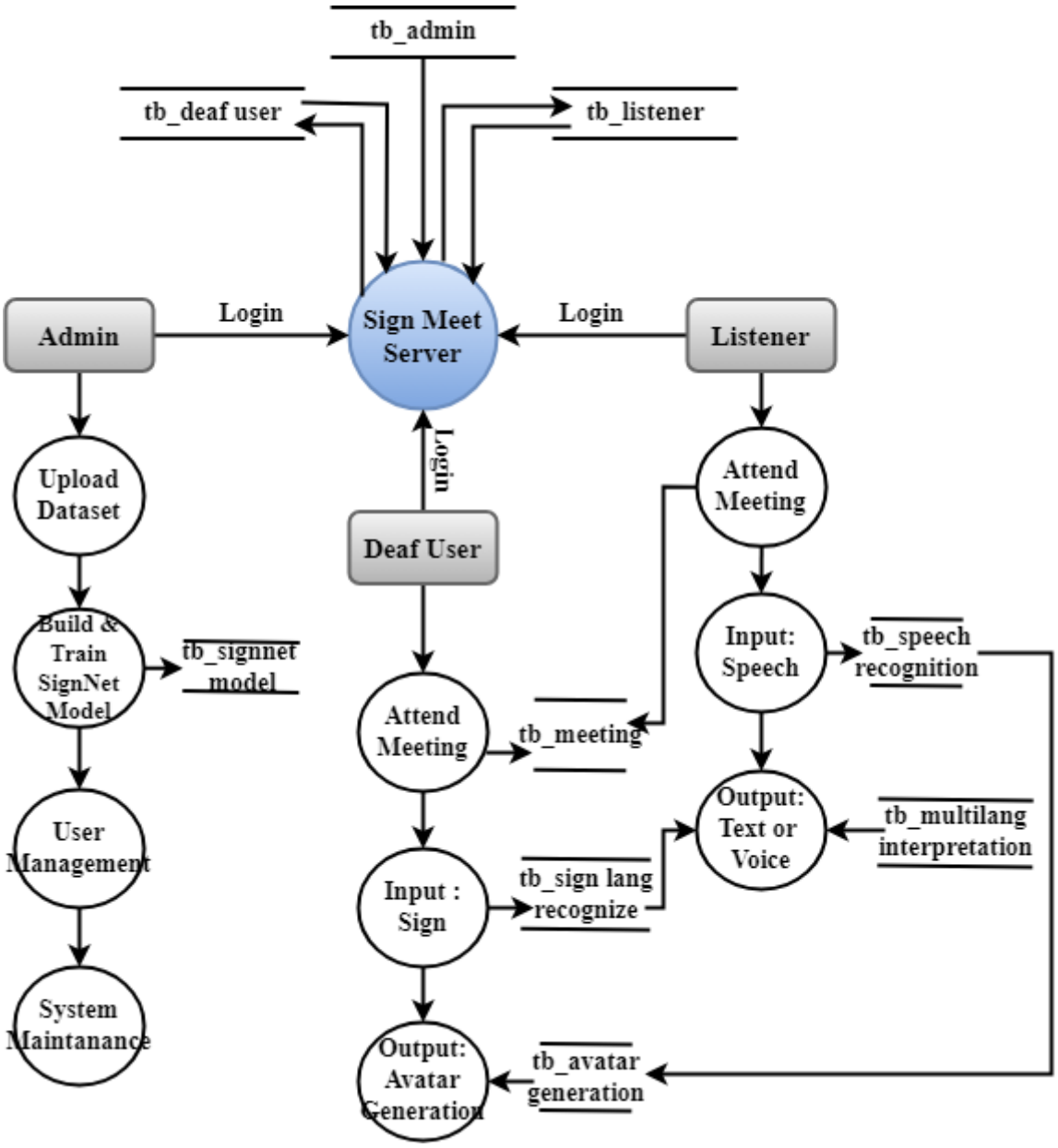



**Fig 6.2.3. Data Flow Level 2**

## 6.3 UML DIAGRAM

## USECASE DIAGRAM

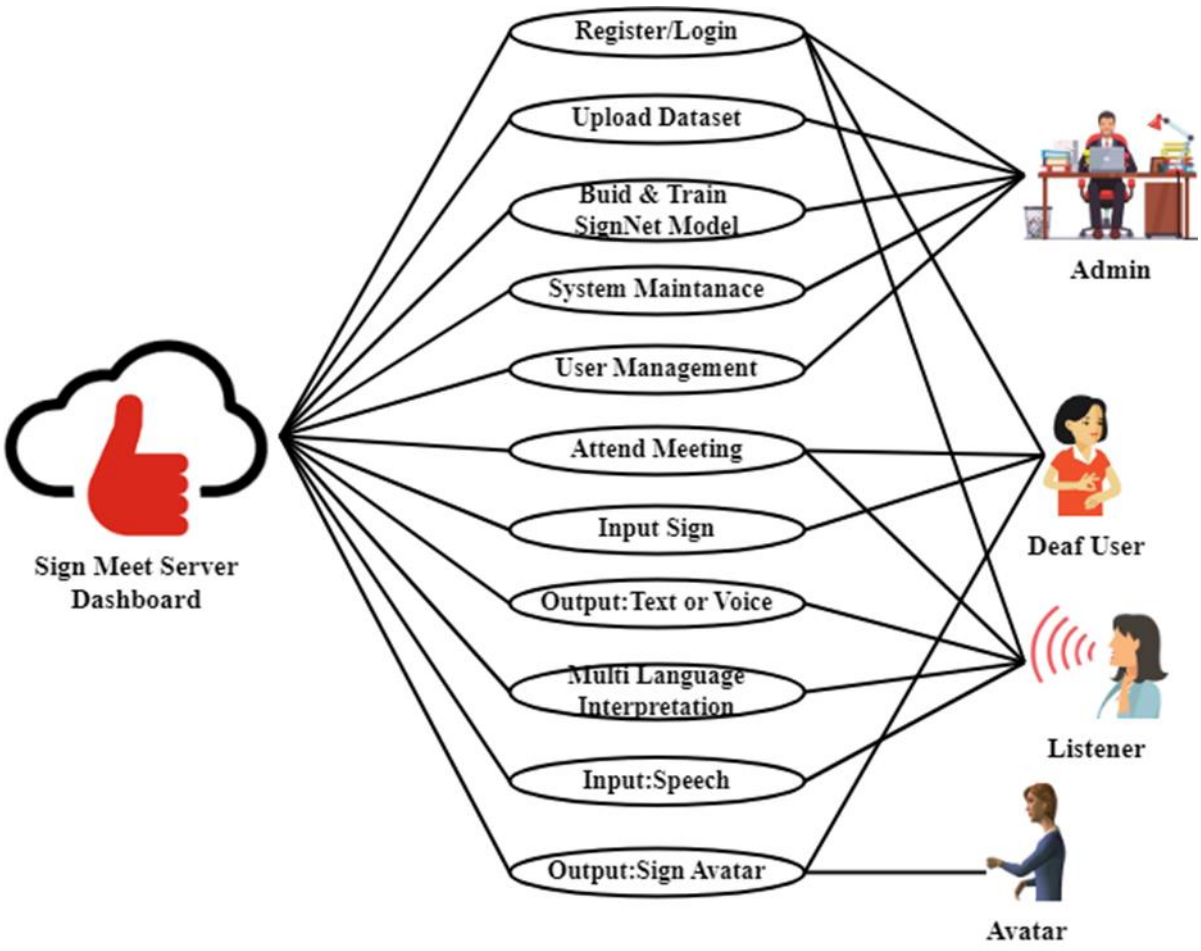



**Fig 6.3.1 Use Case Diagram**

## CLASS DIAGRAM

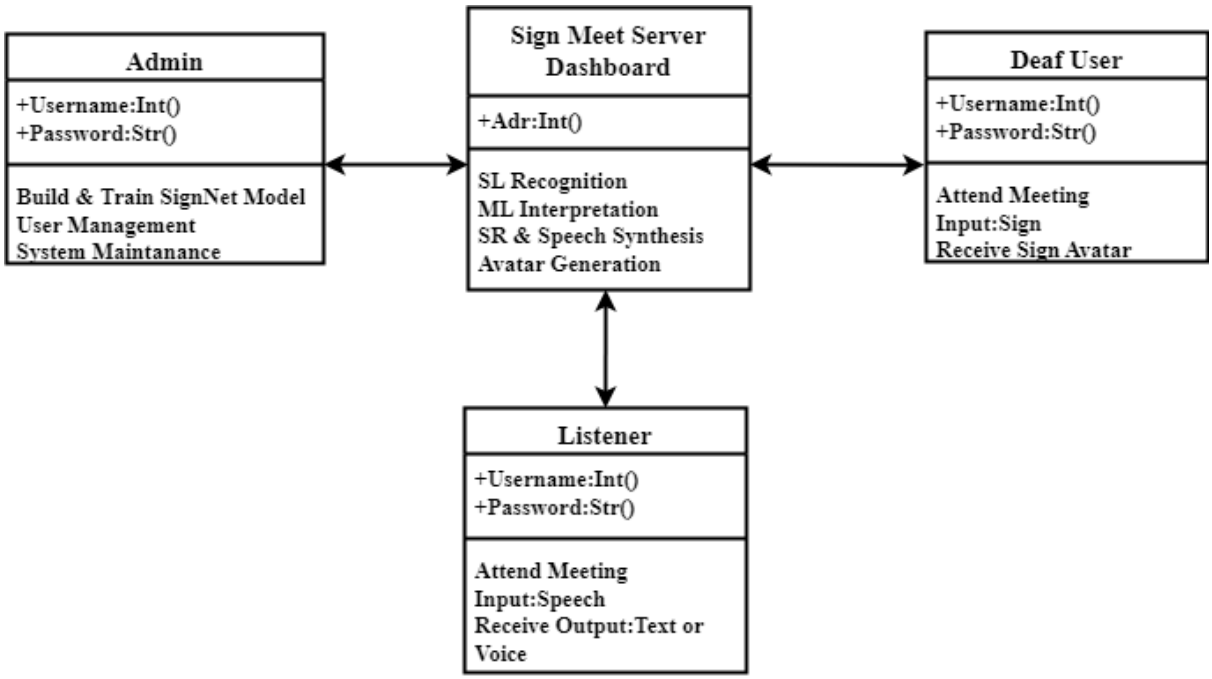



**Fig 6.3.2 Class Diagram**

## ACTIVITY DIAGRAM

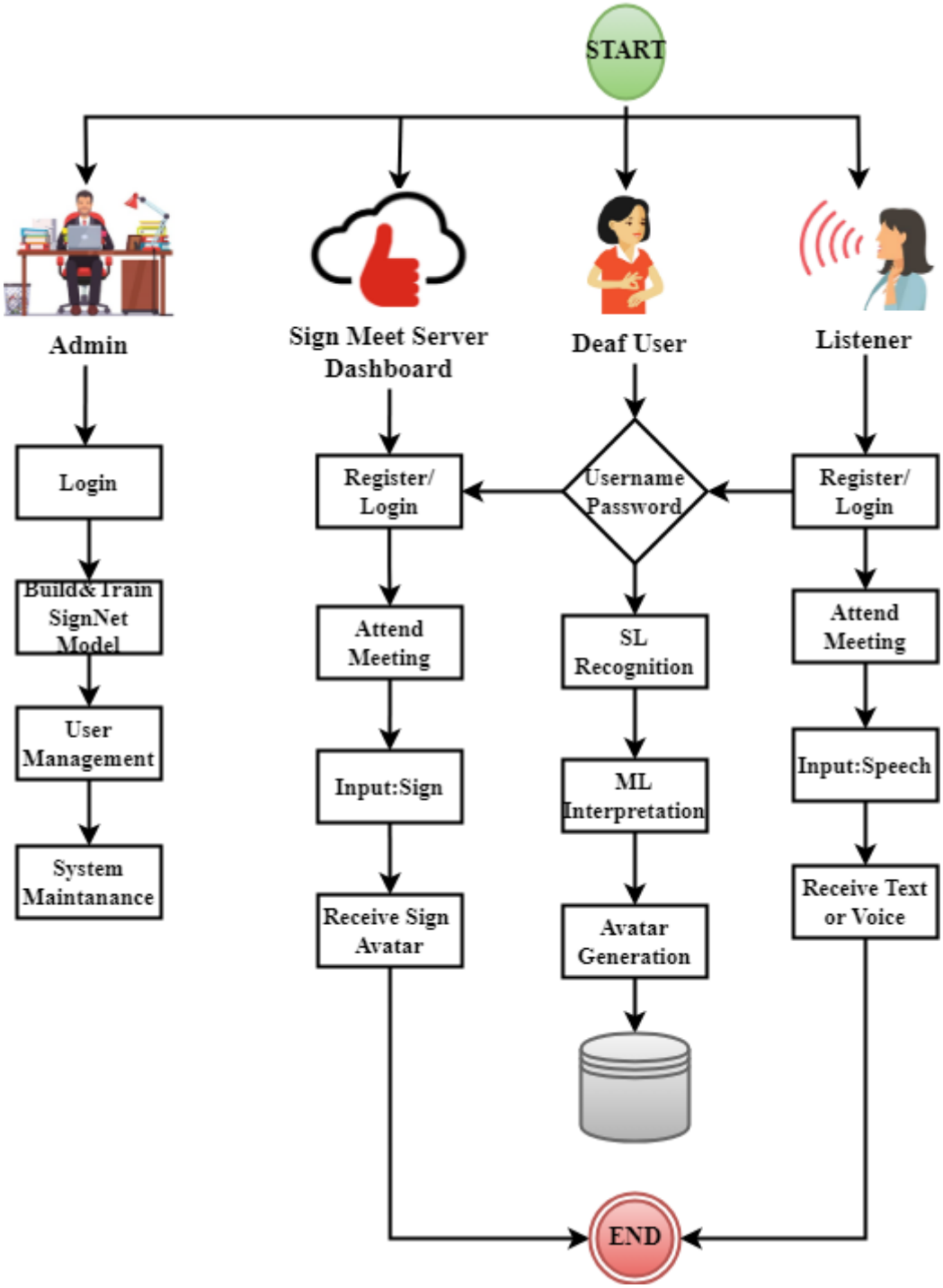



**Fig 6.3.3 Activity Diagram**

## SEQUENCE DIAGRAM

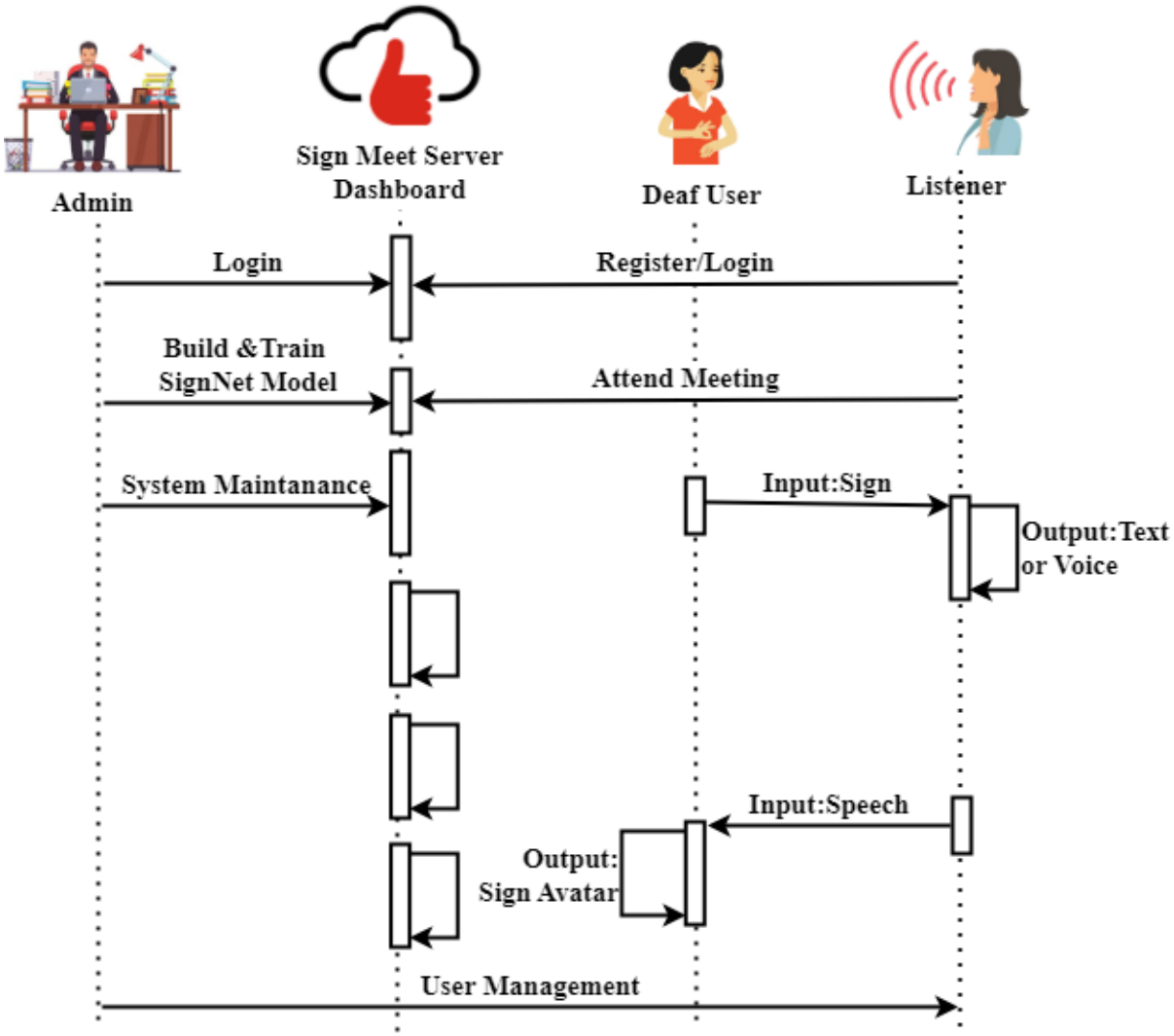



**Fig 6.3.4 Sequence Diagram**

## COLLABORATION DIAGRAM

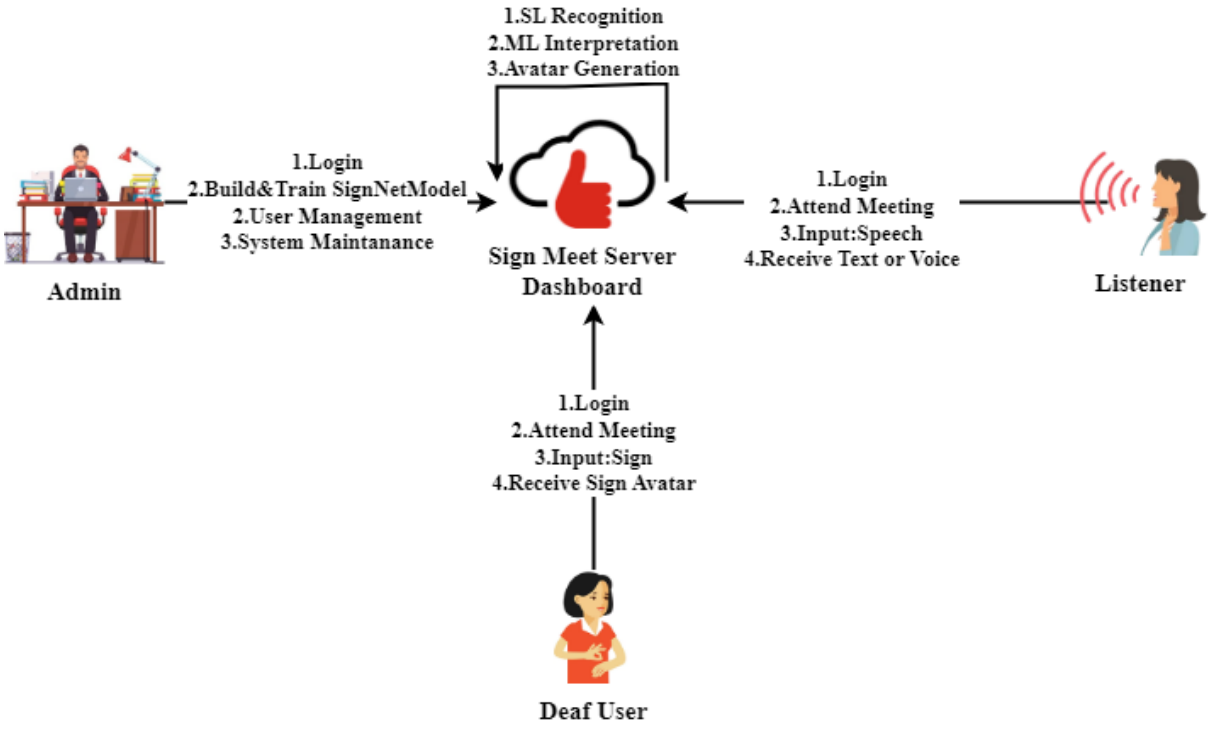



**Fig 6.3.5 Collaboration Diagram**

## COMPONENT DIAGRAM

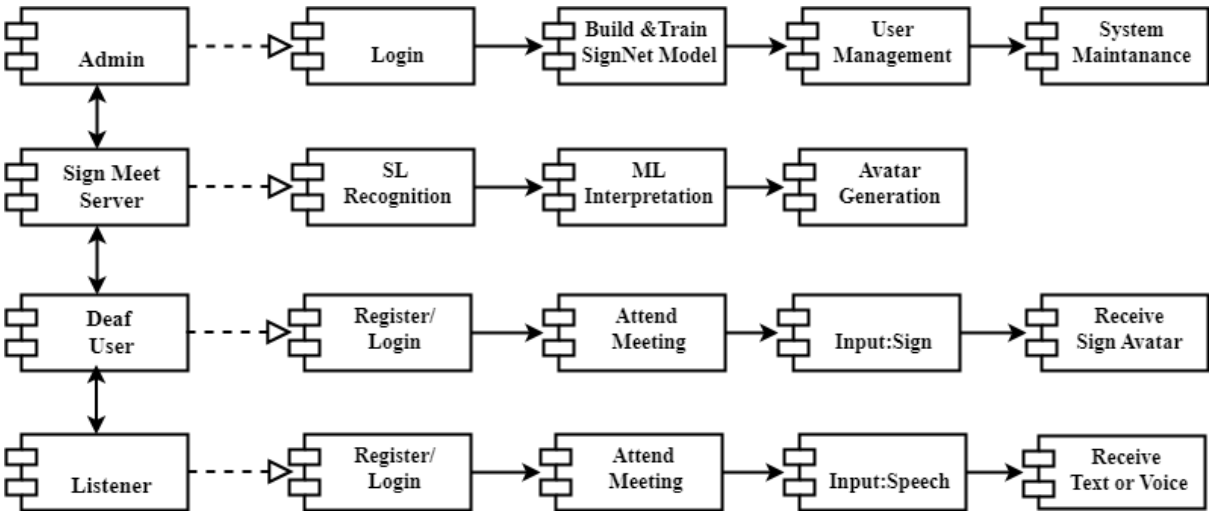



**Fig 6.3.6. component Diagram**

# CHAPTER 7

# SYSTEM IMPLEMENTATION

System implementation involves deploying the developed components into a functional system, ensuring smooth operation and usability. This phase includes integrating various modules, testing, and deploying the system in a real-time environment.

## 1. Development and Integration

The system is developed using Python, Flask, MySQL, Bootstrap, and WampServer. Each module, including sign language recognition, speech synthesis, and avatar generation, is implemented separately and then integrated into the **Sign Meet Web App**, ensuring seamless communication between deaf and non-deaf users.

## 2. Sign Language Recognition Deployment

The **SignNet Model** is trained using Temporal Convolutional Networks (TCNs) and deployed into the web application. The system processes live video input, recognizes sign language gestures, and converts them into text or speech.

## 3. Speech Recognition and Synthesis

Speech input is preprocessed using Wavelet Transform, and feature extraction is done using MFCC. Hidden Markov Models (HMMs) are used for speech recognition, converting spoken words into text, which is then translated into sign language through an animated avatar.

## 4. Avatar Integration for Visual Communication

A real-time **Avatar Module** is implemented to interpret spoken words and convert them into sign language gestures, enhancing communication accessibility.

### 5. Virtual Meeting Integration

The system is integrated with Jitsi virtual platforms, enabling real-time translation, sign recognition, and multi-language interpretation in online meetings.

### 6. Database and Notification System

A **MySQL database** is implemented to manage users (admin, deaf users, non-deaf users), meeting logs, and sign recognition data. A notification system is also deployed to share meeting links via email and SMS.

### 7. Testing and Deployment

Rigorous testing is conducted, including unit testing, integration testing, and user acceptance testing. The system is then deployed for real-time use, ensuring inclusivity and accessibility for all users in virtual meetings.

## 7.1. SYSTEM DESCRIPTION

In today's digital era, virtual communication platforms such as Zoom, Microsoft Teams, and Google Meet play a crucial role in professional, educational, and social interactions. However, individuals with hearing and speech impairments often face barriers due to the lack of integrated sign language interpretation features in these platforms. To address this inclusivity gap, the proposed project introduces "SignMeet," an AI-powered system designed to provide real-time bidirectional translation between sign language and spoken language, ensuring that individuals with communication impairments can participate effectively in virtual meetings. The system architecture is composed of four main modules, each contributing to seamless, real-time communication:

- **Sign Recognition Module (SRM):** This module captures video input from the user's camera and utilizes Temporal Convolutional Networks (TCNs) to identify and interpret sign language gestures. The use of TCNs allows

the system to understand temporal dependencies between frames, ensuring accurate recognition of continuous sign language sequences.

- **Speech Recognition and Synthesis Module (SRSM):** Built using Hidden Markov Models (HMMs), this module performs two key functions: converting speech input into text (speech recognition), and generating spoken output from textual content (speech synthesis). This module enables two-way communication by translating spoken words into text for signers and vice versa.
- **Avatar Module (AM):** The avatar module provides a visual, animated representation of spoken language. Text derived from either the speech recognition module or typed input is translated into sign gestures and displayed using a 3D avatar. This ensures that sign language users can receive information in their preferred mode of communication.
- **Platform Integration Module**: To ensure practical usability, the system is designed for seamless integration with major virtual meeting platforms like Zoom, Microsoft Teams, and Google Meet. This integration ensures that signers and non-signers can interact within the same digital environment without needing external tools or switching platforms.

By combining deep learning, computer vision, and natural language processing, the proposed system delivers an end-to-end inclusive communication solution. It significantly enhances the digital accessibility of virtual platforms for the hearing and speech-impaired community, promoting equal participation and reducing communication barriers. The modular design allows for scalability, cross-platform deployment, and potential extension into multilingual and domain-specific applications. In addition, the system's modular and flexible architecture allows for easy updates and incorporation of emerging technologies, such as transformer-based language models and more advanced gesture recognition techniques.

## 7.2. MODULE DESCRIPTION

### 1. Sign Meet Web App

The Sign Meet Web App is a virtual meeting platform designed to bridge the communication gap between deaf and non-deaf users by integrating sign language recognition, speech synthesis, and avatar-based interpretation. The platform is developed using Python, Flask, MySQL, Bootstrap, WampServer, and other necessary libraries and is integrated with Jitsi, an open-source virtual meeting platform, for real-time video conferencing. This system ensures that users with hearing or speech impairments can actively participate in online meetings, lectures, and discussions without barriers. the Sign Meet Web App provides a comprehensive, AI-powered solution for enabling inclusive virtual communication. By integrating Jitsi for video conferencing and leveraging advanced AI techniques. ensuring a more accessible and connected virtual environment for everyone.

### 2. End User Interface

The End User Module defines the different roles within the Sign Meet Web App, ensuring that each user has specific functionalities tailored to their needs. The system supports three main types of users: Admin, Deaf Users, and Non-Deaf Users, each playing a crucial role in the platform's functionality.

### 2.1 Admin Module

The Admin Module is responsible for managing the platform and ensuring smooth operations. Administrators have access to critical functionalities such as user management, model training, and system monitoring.

- **Login & Authentication**: The admin logs in using secure credentials, gaining access to the system's backend.

- **User Management**: Admins can register, approve, or manage accounts for both deaf and non-deaf users, ensuring that only authorized individuals can use the platform.
- **Train the SignNet Model**: The admin is responsible for training and updating the **SignNet Model**, which is used for **real-time sign language recognition**. This involves feeding new datasets, optimizing the model, and ensuring high accuracy in recognizing gestures.
- **Monitor System Performance**: Admins track system usage, analyze logs, and address any technical issues to maintain seamless functionality.

The admin module ensures that the **Sign Meet Web App** remains efficient, up-to-date, and accessible to all users.

### 2.2 Deaf User Module

The Deaf User Module is designed for users with hearing impairments, providing them with the necessary tools to communicate effectively in virtual meetings. This module focuses on registration, meeting creation, and interaction through sign language recognition.

### 3. SignNet Model: Build and Train

The SignNet Model is the core AI-powered sign language recognition system used in the Sign Meet Web App. This model is responsible for real-time gesture recognition, processing live video input, extracting key features, and classifying hand gestures into meaningful sign language representations.

It is trained using advanced deep learning techniques to ensure high accuracy in interpreting sign language. The model undergoes multiple stages, including preprocessing, feature extraction, classification, and deployment, to provide seamless interaction. It contains the CNN and TCN deep learning algorithms such as random forest for faster and better training of the model.

### 3.1 Live Video Input

The SignNet Model starts by capturing live video from the user during a virtual meeting. This video feed contains sign language gestures that need to be processed and translated into text or speech.

- The system accesses the user's webcam or device camera to capture hand and facial gestures.
- The video input is processed in real-time, ensuring minimal latency for smooth communication.
- Frame-by-frame data is extracted for further processing.

### 3.2 Preprocessing

Preprocessing is essential for cleaning and refining the captured video frames before feeding them into the SignNet Model. Several techniques are applied to enhance the input data:

- **Frame Conversion**: Extracts individual frames from the live video stream.
- **RGB to Grayscale Conversion**: Reduces computational complexity by converting color images into grayscale.
- **Resize**: Ensures uniform image dimensions for consistent model training and testing.
- **Noise Filtering using Gabor Filter**: Removes unwanted noise and enhances important features like hand edges.
- **Binarization**: Converts images into black-and-white representations, simplifying hand shape detection.
- **Segmentation using RPN (Region Proposal Network)**: Detects and isolates hand gestures from the background, making it easier for the model to focus on relevant features. This preprocessing stage optimizes the quality of input data and enhances recognition accuracy. And gets skeleton data for better processing of gestures.

## 4. Sign Language Recognition

The Sign Language Recognition module is a critical component of the Sign Meet Web App, enabling seamless communication between deaf and non-deaf users. This module is designed to process live video input and recognize sign language gestures in real time, translating them into text or speech output. It utilizes Temporal Convolutional Networks (TCN) in conjunction with the trained SignNet Model to achieve high accuracy in sign language recognition.

### 4.1 Live Video with Sign

The recognition process begins with the live video input from the user's camera. The system continuously captures video frames to track hand movements, facial expressions, and body gestures relevant to sign language. Advanced motion tracking techniques ensure that the system accurately follows the user's gestures, even in varying lighting conditions and different backgrounds. To enhance the accuracy of recognition, the video stream undergoes preprocessing. This involves converting the video frames into grayscale to reduce complexity, resizing them for uniformity, and applying noise filtering techniques such as the Gabor filter to remove unnecessary background distractions. Segmentation using a Region Proposal Network (RPN) ensures that only the relevant hand gestures are isolated, allowing for precise recognition.

### 4.2 Recognizing Sign

After preprocessing and feature extraction, the system utilizes Temporal Convolutional Networks (TCN) to recognize the signs in real time. Unlike traditional classification methods, TCN is designed to process sequential data, making it highly effective for gesture-based recognition. Instead of analyzing a single frame, TCN considers the entire sequence of movements, allowing it to accurately distinguish between similar gestures.

### 5. Multilanguage Interpretation

The Multilanguage Interpretation module enables real-time translation of sign language into multiple spoken and written languages, ensuring seamless communication for deaf and non-deaf users. It leverages Natural Language Processing (NLP) and deep learning to accurately translate recognized signs into text or speech across various languages. The system also supports speech-to-sign language conversion by using speech recognition techniques like MFCC and HMMs to transcribe spoken words into text, which is then converted into sign language and displayed via an AI-powered avatar. It adapts to regional variations and continuously improves through user interactions.

### 6. Speech Recognition

The Speech Recognition and Visual Communication module facilitates effective interaction between non-deaf and deaf users by converting spoken language into text and sign language. This enables non-deaf users to ask questions and receive responses in a visually accessible format.The speech recognition process begins with preprocessing using Wavelet Transform, which enhances speech clarity by filtering out noise. Next, Mel-Frequency Cepstral Coefficients (MFCC) are used for feature extraction, capturing essential speech characteristics to improve recognition accuracy. The system then utilizes Hidden Markov Models (HMMs) for speech-to-text conversion, ensuring accurate interpretation of spoken words.

### 7. Avatar Generation

The Avatar Generation module plays a crucial role in bridging the communication gap between deaf and non-deaf users by providing a real-time, animated visual representation of spoken language. This module is designed to convert recognized speech or text into sign language gestures using an AI-powered avatar, ensuring an inclusive and accessible communication experience.

# CHAPTER 8
# TESTING

## 8.1. SOFTWARE TESTING

Software testing is the process of evaluating and verifying that a software application functions correctly, meets specified requirements, and is free of defects. It ensures the reliability, security, and performance of the system. Testing is essential to identify bugs, improve functionality, and enhance user experience before deployment. For this project, testing is to ensure accurate sign language recognition, seamless virtual communication, proper speech-to-text conversion, avatar generation, and integration with virtual platforms like Jitsi.

## TYPES OF TESTING

- **Functional Testing**

Functional testing verifies that the application performs as expected according to the specified requirements. It checks whether the system's features, such as sign language recognition, speech-to-text conversion, avatar generation, and meeting functionalities, work correctly. This ensures that users can communicate without issues in virtual meetings.

- **Unit Testing**

Unit testing focuses on testing individual components of the system to ensure they function correctly before integration. For this project, unit tests will be conducted on modules like the SignNet Model, speech recognition module, and avatar generation module to verify their standalone functionality.

- **Usability Testing**

Usability testing ensures that the system is user-friendly, intuitive, and accessible to both deaf and non-deaf users. It assesses whether users can easily navigate the platform, join meetings, recognize signs, and interact with the avatar-based sign language interpreter. Feedback from test users helps refine the interface for better accessibility.

## 8.2. TEST CASES

Test cases are a set of predefined conditions used to verify the functionality, reliability, and accuracy of the project. Each test case outlines specific inputs, expected outputs, and actual results to ensure the system meets its requirements. By executing these test cases, potential defects can be identified and resolved, enhancing the overall performance and user experience of the application. And gives understanding o the issues that was or will persist in the software and performing the needed solutions to resolve the issue. Additionally, test cases serve as a foundation for quality assurance by systematically validating each feature and functionality under various scenarios, including edge cases and unexpected user behaviour.

1. **Test Case ID:** TC001
   - **Input:** Admin enters valid credentials and clicks "Login"
   - **Expected Result:** Admin dashboard should be displayed
   - **Actual Result:** Admin dashboard displayed successfully
   - **Status:** Passed

2. **Test Case ID:** TC002
   - **Input:** Admin enters invalid credentials and clicks "Login"
   - **Expected Result:** Error message "Invalid username or password" should be displayed
   - **Actual Result:** Error message displayed
   - **Status:** Passed

3. **Test Case ID:** TC003
   - **Input:** Deaf user registers with all required details
   - **Expected Result:** Account should be created successfully
   - **Actual Result:** Account created successfully
   - **Status:** Passed

## 8.3. TEST REPORT

### Introduction

The test report documents the testing process carried out for the project is to ensure its accuracy, efficiency, and seamless integration with virtual meeting platforms. It includes test objectives, scope, environment, results, and identified bugs.

### Test Scope

The testing scope covers the Sign Meet Web App modules, including user authentication, real-time sign language recognition, speech-to-text conversion, avatar display, and system integration with virtual platforms like Jitsi.

### Test Objective

The objective of this testing is to validate the proper functionality of sign language recognition, speech synthesis, avatar generation, and multilingual interpretation within the virtual meeting environment.

### Test Result

The system has been tested for various functionalities, including user authentication, sign recognition accuracy, speech synthesis, and avatar generation.

### Bug Report

A bug report records issues, defects, or errors encountered during testing. It includes details such as Bug ID, Test Case ID, Bug Description, Bug Status, and Output, ensuring developers can track and fix the issues efficiently.

### Test Conclusion

The project has undergone rigorous testing, and most functionalities performed as expected. Identified bugs are documented and will be addressed in the next development cycle to enhance system performance and usability.

# CHAPTER 9

# PERFORMANCE ANALYSIS

## Confusion Matrix

The confusion matrix provides a clear overview of how well the model performs on sign recognition by comparing actual signs with predicted ones:

- **True Positives (TP):** Signs correctly identified as the intended gesture.
- **False Positives (FP):** Signs incorrectly predicted as a gesture when they are not.
- **True Negatives (TN):** Correct rejections of non-gesture inputs.
- **False Negatives (FN):** Signs that were not recognized by the system.

This matrix is key for understanding the model's behavior in real-time recognition and helps in tuning the system.

## Accuracy

Accuracy measures the total proportion of correct predictions (both positive and negative) made by the system.

**Formula:**

**Accuracy = (TP + TN) / (TP + TN + FP + FN)**

A high accuracy reflects that the system makes minimal errors across all gesture classes.

## Precision

**Precision** evaluates how many of the positively identified signs were actually correct.

**Formula:**

**Precision = TP / (TP + FP)**

Higher precision ensures the model doesn’t frequently misclassify unrelated movements as valid gestures.

## Recall (Sensitivity)

**Recall** measures the model’s ability to detect all relevant signs.

**Formula:**

**Recall = TP / (TP + FN)**

A high recall indicates that the model misses very few valid gestures during communication.

**F1-Score**

**F1-Score** provides a balance between precision and recall.

**Formula: F1-Score = 2 * (Precision * Recall) / (Precision + Recall)**

It is particularly useful when the dataset is imbalanced or when both false positives and false negatives are critical.

## RESULTS AND DISCUSSION

The performance of the sign recognition system was evaluated using key classification metrics derived from the confusion matrix: accuracy, precision, recall, and F1-score. The confusion matrix offered a detailed visualization of the model's ability to correctly identify gesture signs versus incorrect or missed classifications. The system demonstrated a high number of True Positives (TP), indicating its strong capability in correctly identifying intended signs. Minimal False Positives (FP) and False Negatives (FN) suggested that the model rarely misclassified non-gesture inputs or missed actual gestures.

The results indicate that the sign recognition system effectively identifies and classifies hand gestures with high precision and recall. The use of a confusion matrix provided crucial insights into the model's strengths and weaknesses, aiding in performance analysis and system optimization. The high accuracy reflects the robustness of the underlying model in differentiating between valid gestures and irrelevant movements. Precision was particularly significant in preventing the model from reacting to unintended or noisy inputs, enhancing reliability in real-world settings.

# CHAPTER 10

# CONCLUSION AND FUTURE SCOPE

In conclusion, this project successfully bridges the communication gap between deaf and non-deaf users by integrating advanced Sign Language Recognition, Speech Recognition, and Avatar-Based Visual Communication within a virtual meeting platform. The module effectively recognizes sign language gestures using the SignNet Model trained with Temporal Convolutional Networks (TCNs) and converts speech into sign language using Hidden Markov Models (HMMs) and an animated avatar. By integrating with Jitsi virtual platforms, the module ensures seamless real-time communication in online meetings. Additionally, the multilanguage interpretation, speech synthesis, and notification system enhance accessibility and usability for a diverse group of users. This module provides a scalable and inclusive communication platform, significantly improving virtual interactions for the deaf community. This project provides a scalable and inclusive communication platform, significantly improving virtual interactions for the deaf community. Future enhancements could include expanded language support, AI-driven sign recognition improvements, and deeper integration with other conferencing platforms, further advancing accessibility and inclusivity. Building upon the existing systems and addressing their limitations—such as the unavailability of real-time interpretation, reliance on costly human interpreters, and lack of contextual understanding—this project offers a much-needed technological upgrade. Unlike traditional assistive devices or pre-recorded videos, the proposed solution delivers dynamic, real-time interaction through AI-powered recognition and translation. By leveraging cutting-edge AI technologies and focusing on user-centric design, this project not only enhances communication but also promotes digital equity. It empowers the deaf community to participate fully and independently in professional, educational, and social virtual environments.

This project can be further improved with advanced features and optimizations to enhance accessibility, accuracy, and user experience. Some key future enhancements include:

- **Expanded Language Support**: Incorporating additional sign languages, including regional and country-specific variations, to support a wider audience.
- **Integration with More Virtual Platforms**: Extending compatibility beyond Jitsi to platforms like Zoom, Microsoft Teams, and Google Meet, allowing broader adoption.
- **Enhanced Avatar Customization**: Improving the Avatar Module with realistic 3D animations, facial expressions, and lip-syncing for better sign language representation.
- **Mobile App Development**: Creating a mobile version of the project to allow accessibility on smartphones and tablets for on-the-go communication.
- **Automated Meeting Transcription and Summarization**: Implementing AI-based summarization tools to provide meeting notes and transcripts for improved accessibility and documentation.

By incorporating these enhancements, the proposed system can evolve into a comprehensive and highly accessible communication tool, fostering a truly inclusive virtual environment. Building on these proposed enhancements, the system can also benefit from continuous learning and personalization features. For instance, integrating machine learning algorithms that adapt to individual user behavior and communication styles can significantly improve the accuracy of sign recognition and speech synthesis. Additionally, incorporating real-time feedback mechanisms would allow users to report translation errors or suggest improvements, contributing to a more refined and user-driven system.

# APPENDIX 1: SAMPLE CODINGS

## Packages

```
from flask import Flask, render_template, Response, redirect, request, session,
abort, url_for
from camera import VideoCamera
from camera1 import VideoCamera1
import mysql.connector
from random import randint
from urllib.request import urlopen
import cv2
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import shutil
import imagehash
from werkzeug.utils import secure_filename
from PIL import Image
import argparse
import urllib.request
import urllib.parse
import pyttsx3
from skimage import transform
from sklearn.model_selection import train_test_split
import joblib
import speech_recognition as sr
from googletrans import Translator
from gtts import gTTS
import mediapipe as mp
```

```
import tensorflow as tf
from PIL import Image
```

**Database Connection**

```
mydb = mysql.connector.connect(
host="localhost",
user="root",
password="",
charset="utf8",
database="sign_meet_new"
```

**Login**

```
def login_user():
msg=""
if request.method=='POST':
uname=request.form['uname']
pwd=request.form['pass']
cursor = mydb.cursor()
cursor.execute('SELECT * FROM sign_user WHERE uname = %s AND pass =
%s', (uname, pwd))
account = cursor.fetchone()
if account:
session['username'] = uname
ff=open("static/msg.txt","w")
ff.write("")
ff.close()
cursor.execute('SELECT count(*) FROM sign_user WHERE status=1')
dd = cursor.fetchone()[0]
if dd==0:
cursor.execute("update sign_user set status=1 where uname=%s",(uname,))
mydb.commit()
```

```
ff=open("static/deaf.txt","w")
ff.write(uname)
ff.close()
return redirect(url_for('test_cam'))
else:
msg = 'Incorrect username/password!'
```

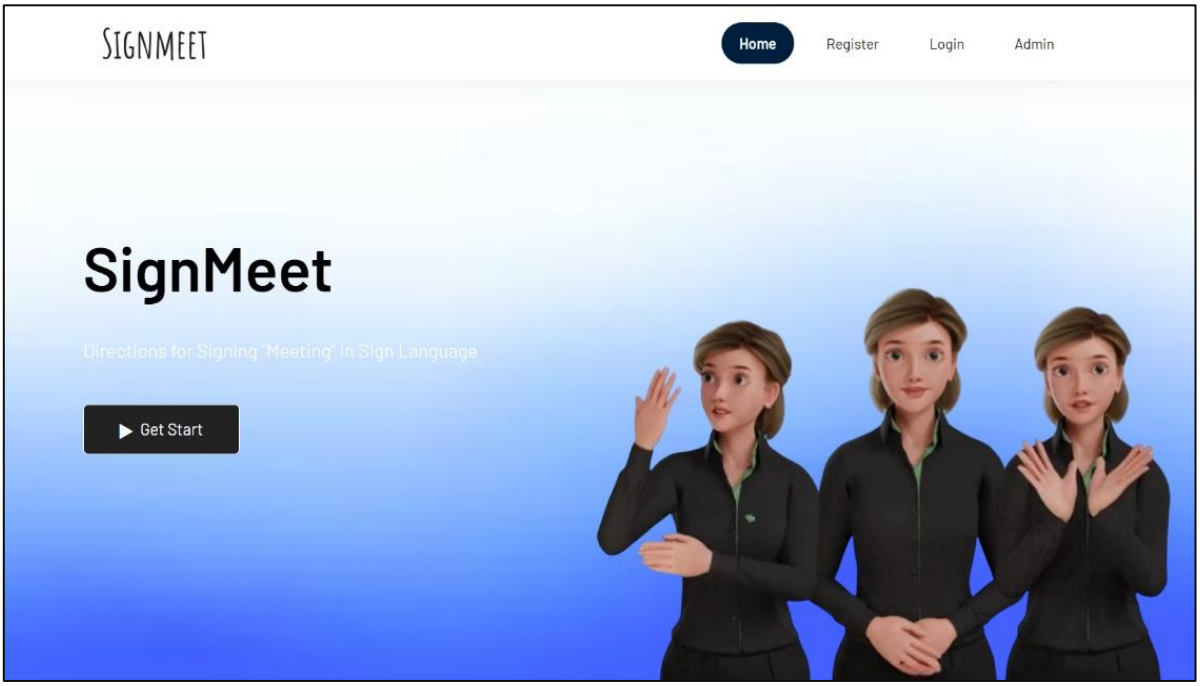
**User Registration**

```
def register():
msg=""
now = datetime.datetime.now()
rdate=now.strftime("%d-%m-%Y")
mycursor = mydb.cursor()
if request.method=='POST':
name=request.form['name']
mobile=request.form['mobile']
email=request.form['email']
uname=request.form['uname']
pass1=request.form['pass']
mycursor.execute('SELECT count(*) FROM sign_user WHERE uname = %s',
(uname,))
cnt = mycursor.fetchone()[0]
if cnt==0:
mycursor.execute("SELECT max(id)+1 FROM sign_user")
maxid = mycursor.fetchone()[0]
if maxid is None:
maxid=1
sql = "INSERT INTO sign_user(id,name,mobile,email,uname,pass) VALUES
(%s, %s, %s, %s, %s, %s)"
val = (maxid,name,mobile,email,uname,pass1)
```

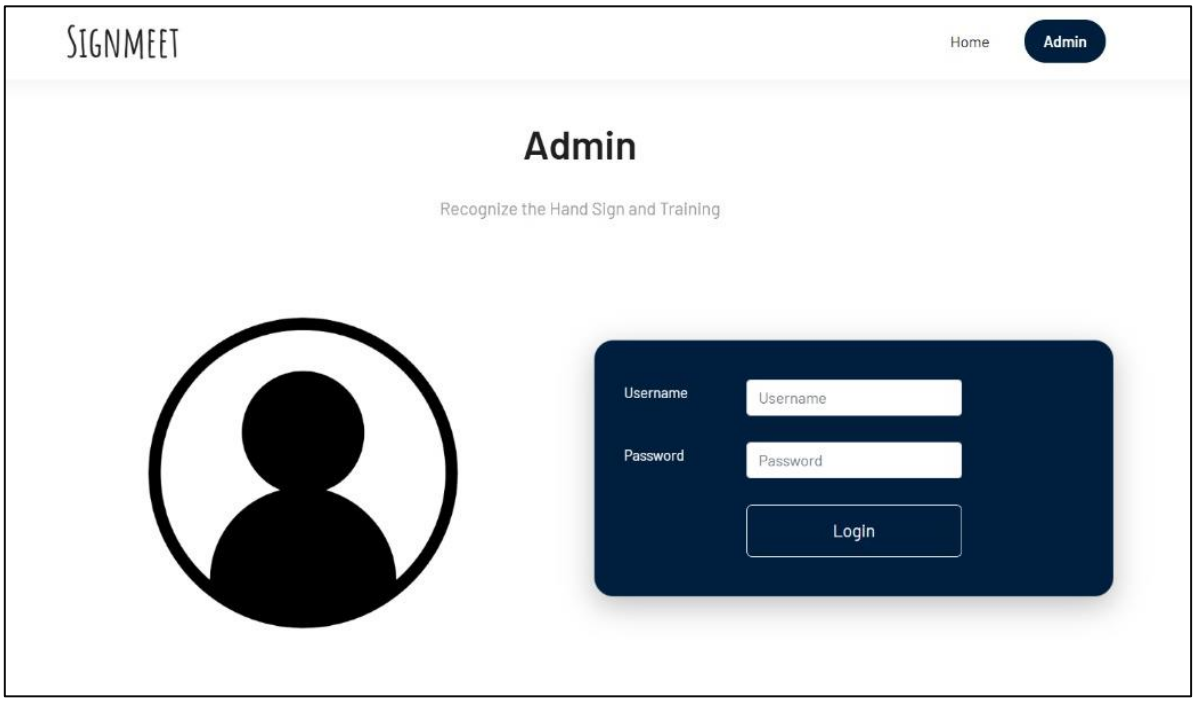
```
from tensorflow_addons.layers import WeightNormalization
# wrap it. WeightNormalization API is different than BatchNormalization or
LayerNormalization.
with K.name_scope('norm_{}'.format(k)):
conv = WeightNormalization(conv)
self._build_layer(conv)
with K.name_scope('norm_{}'.format(k)):
if self.use_batch_norm:
self._build_layer(BatchNormalization())
elif self.use_layer_norm:
self._build_layer(LayerNormalization())
elif self.use_weight_norm:
pass  # done above.
with K.name_scope('act_and_dropout_{}'.format(k)):
self._build_layer(Activation(self.activation,
name='Act_Conv1D_{}'.format(k)))
self._build_layer(SpatialDropout1D(rate=self.dropout_rate,
name='SDropout_{}'.format(k)))
if self.nb_filters != input_shape[-1]:
# 1x1 conv to match the shapes (channel dimension).
name = 'matching_conv1D'
with K.name_scope(name):
# 1x1 conv.
self.shape_match_conv = Conv1D(
filters=self.nb_filters,
kernel_size=1,
padding='same',
return [self.res_output_shape, self.res_output_shape]
```

# APPENDIX 2: SNAP SHOTS

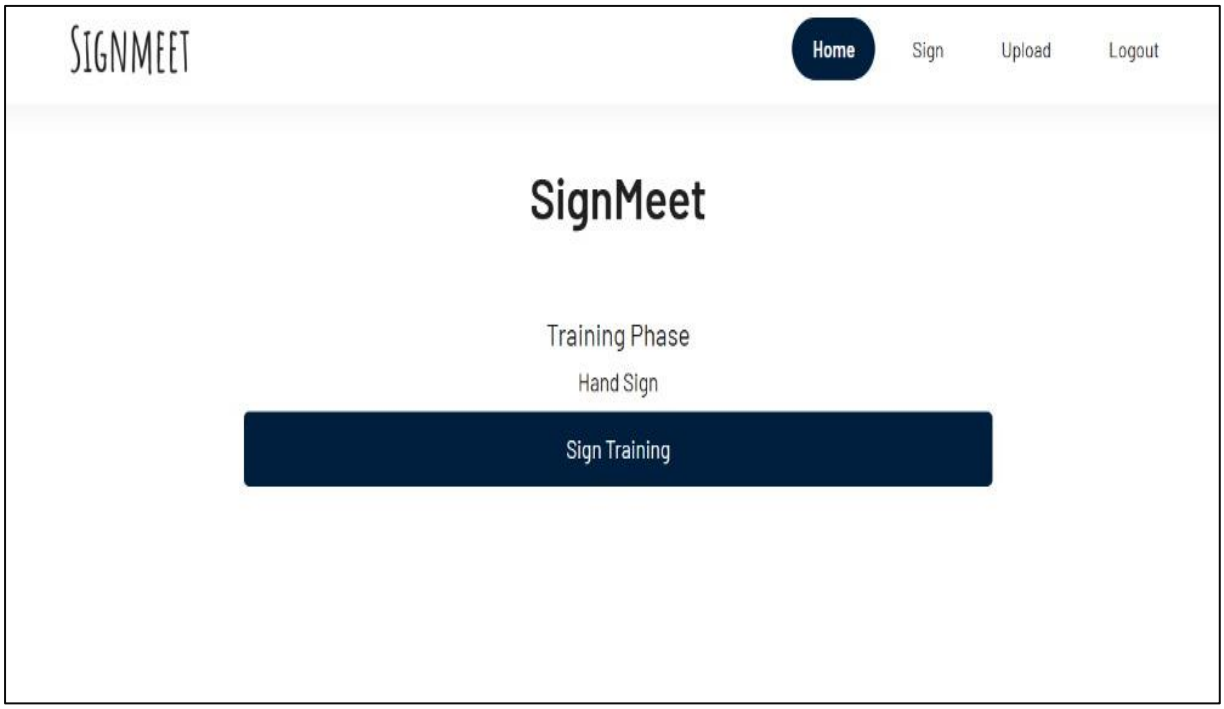
## 1. Admin Dashboard



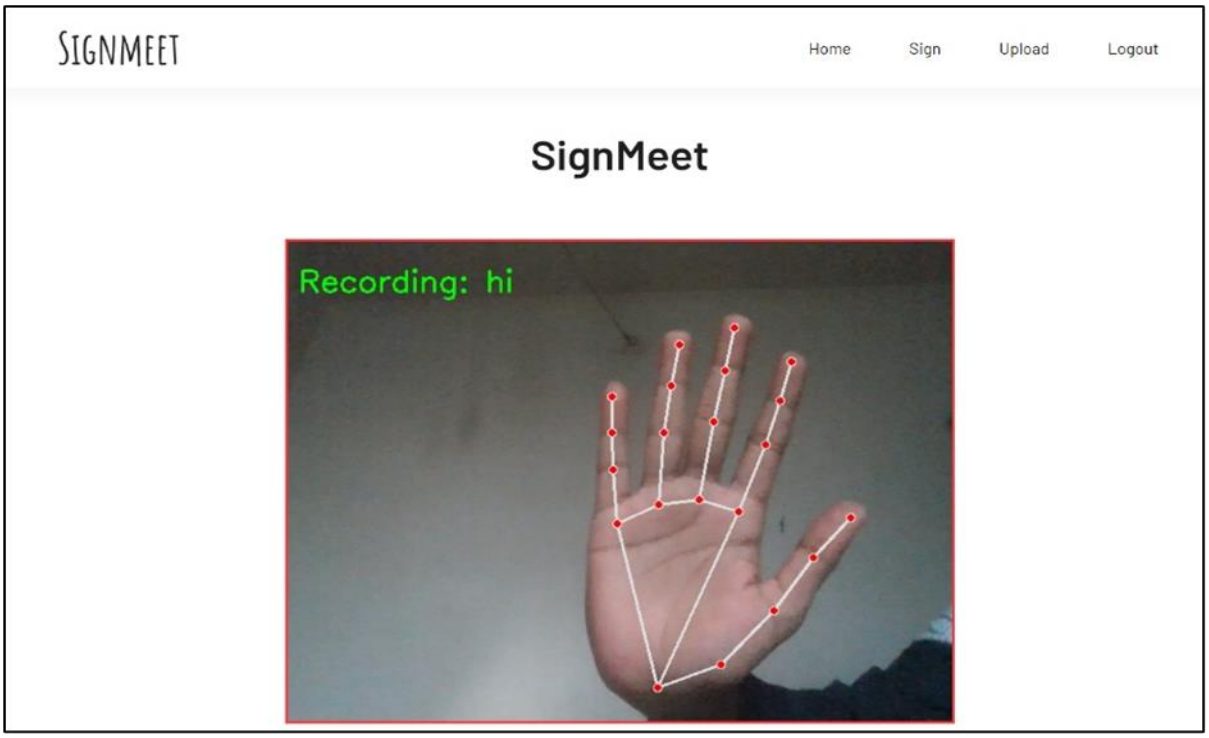

## 2. Admin Login



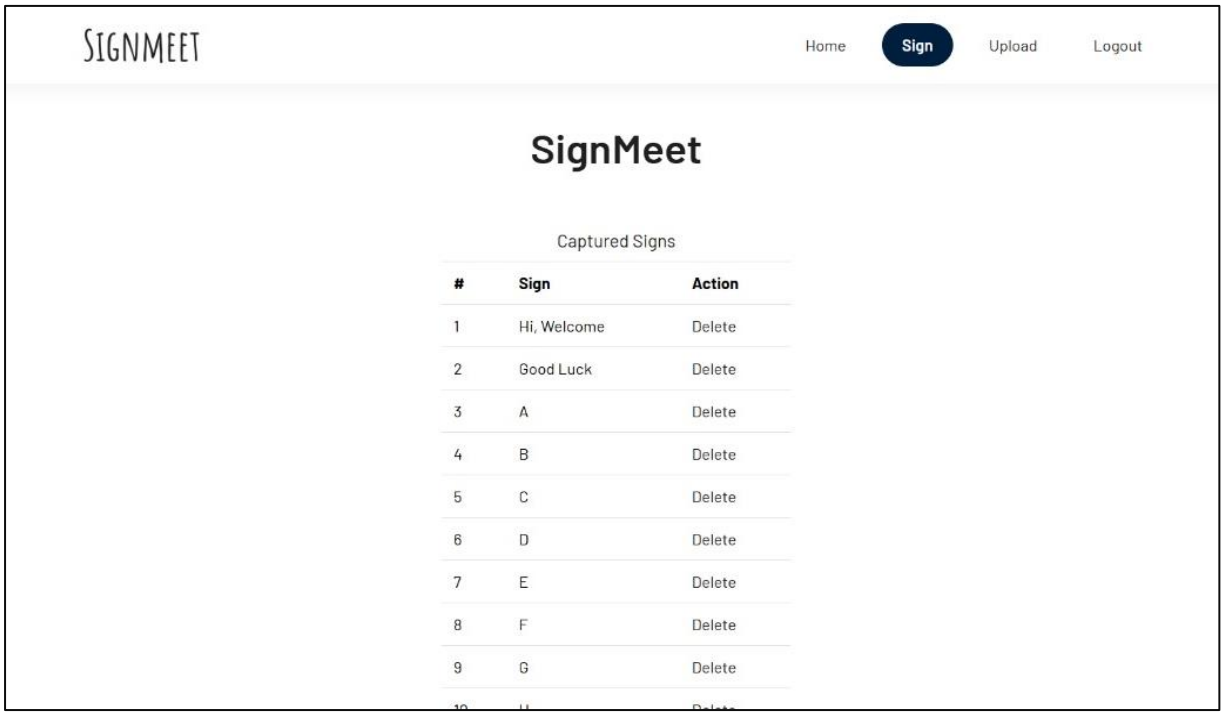
### 3. Training Page



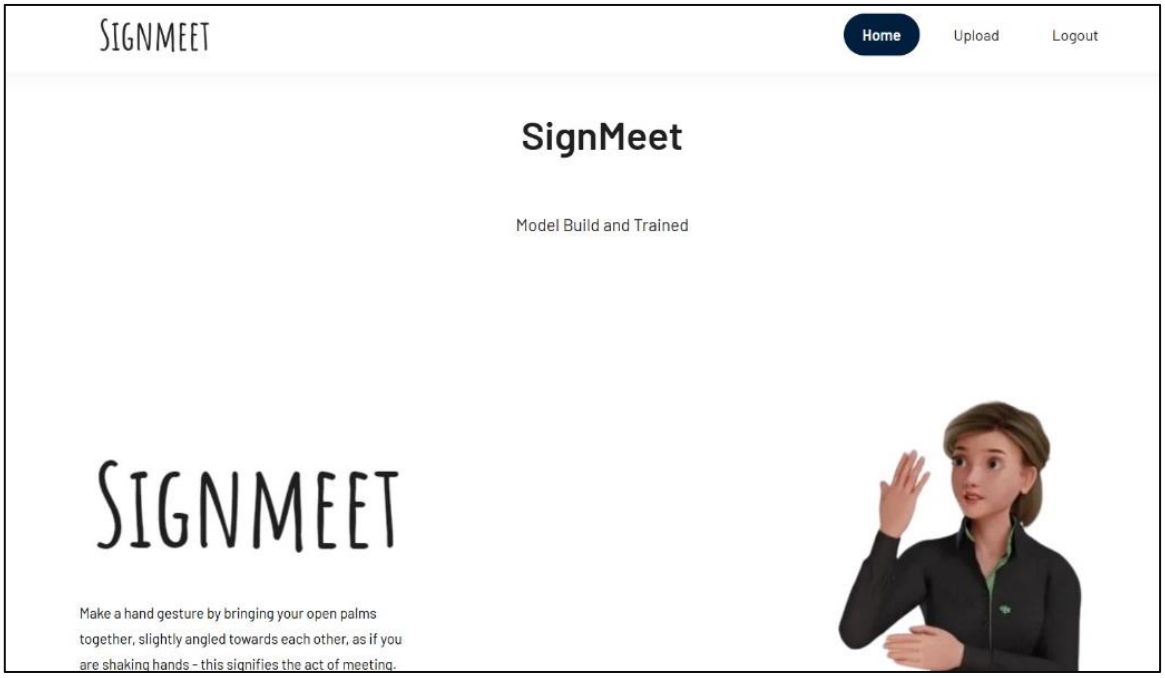

### 4. Hand Sign Detection



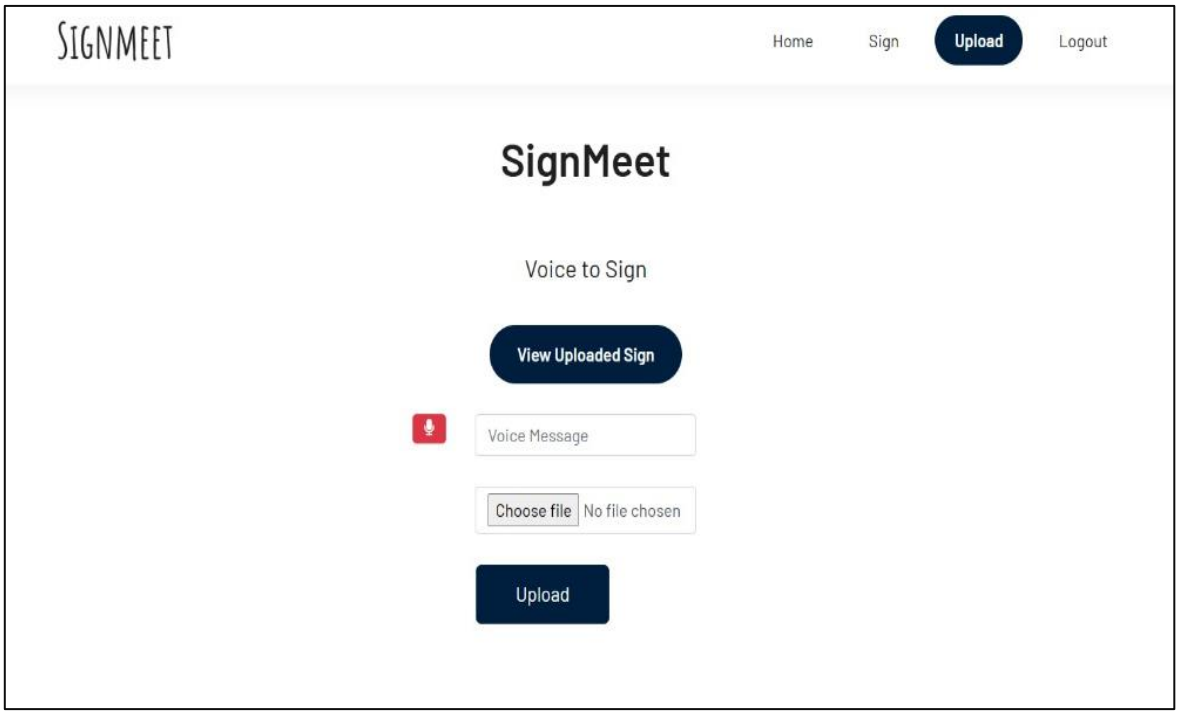
## 5. List of Trained Sign



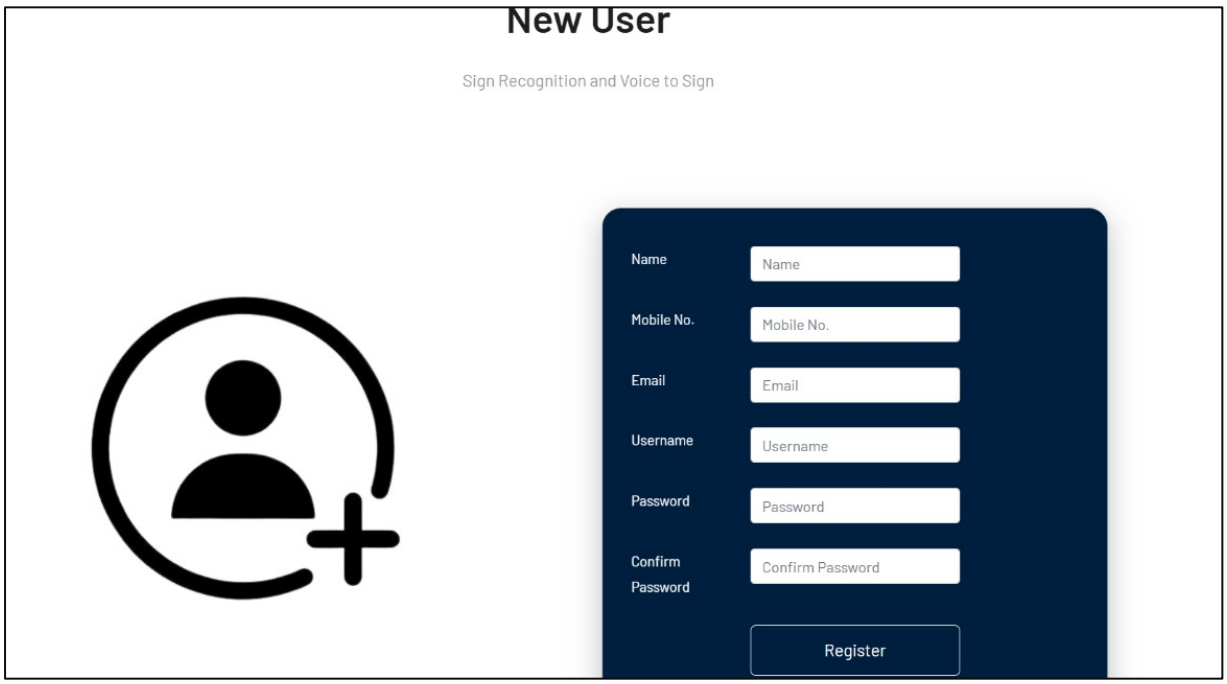

## 6. Training the Model



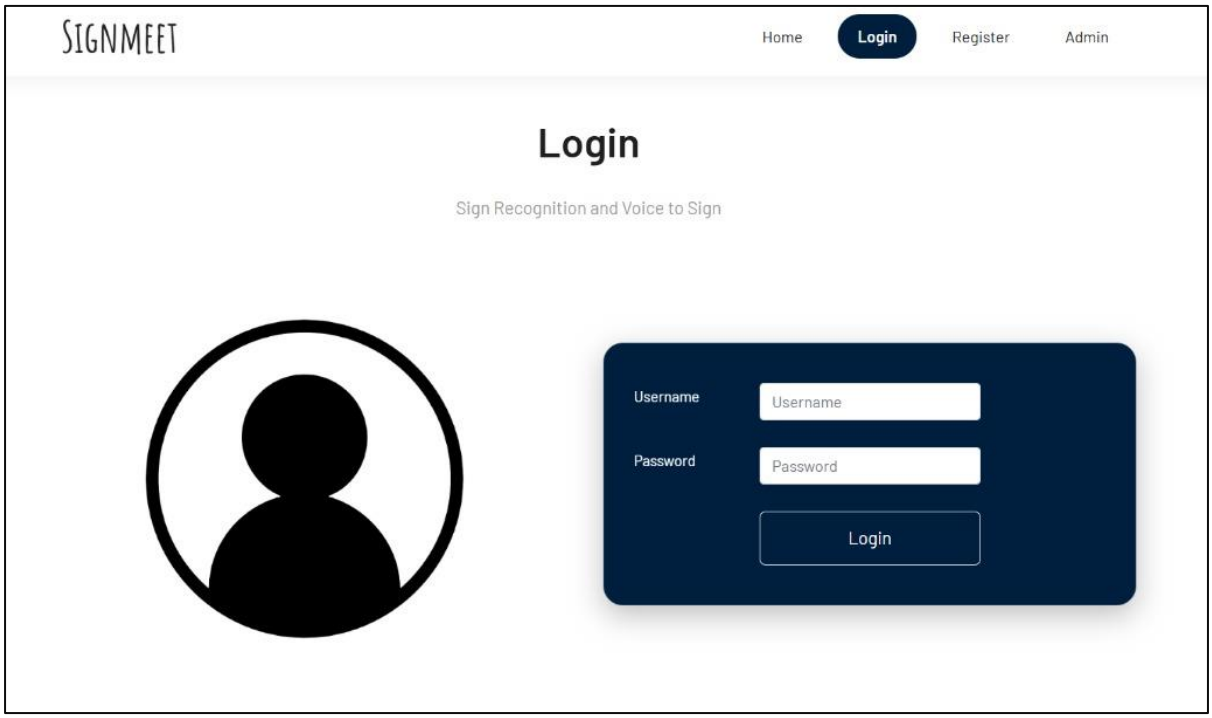
## 7. Voice to Sign



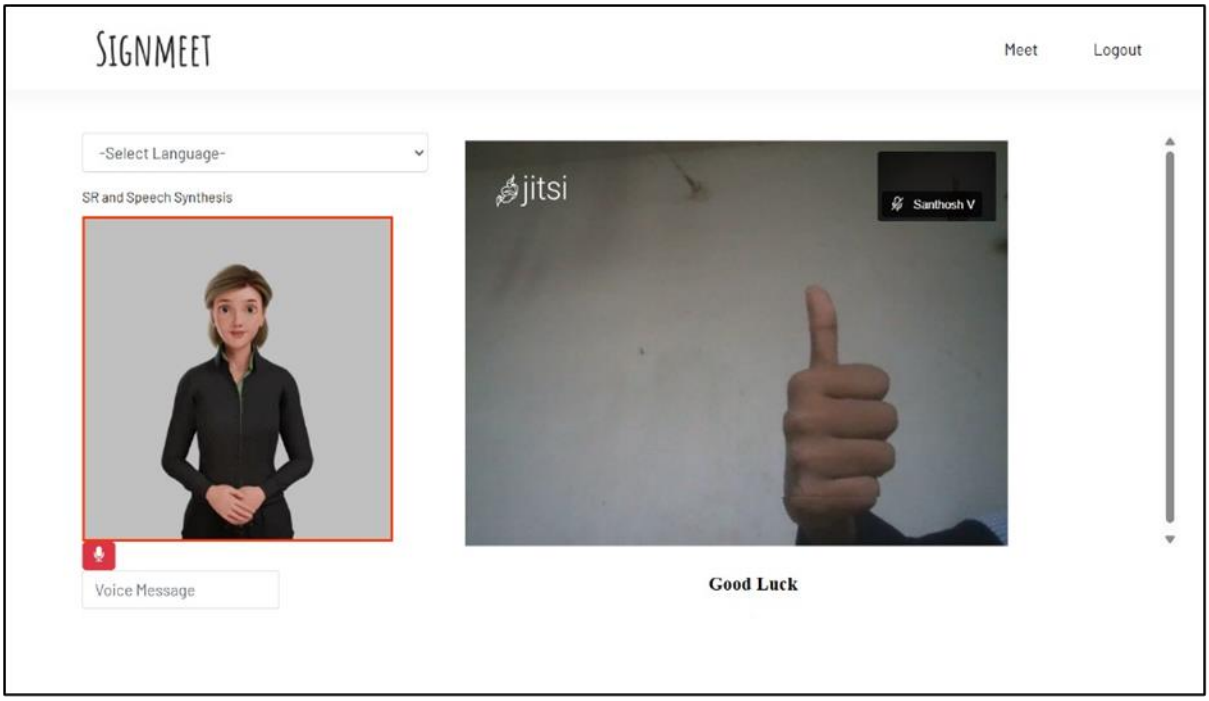

## 8. User Registration



## 9. User Login




## 10. SignMeet Video Conference



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